

ID804

RGB LED Driver IC for Smart Ambient Light

General Description

This device is designed to intelligently respond to the sophisticated ambient lighting scenarios in evolving automotive interiors.

It incorporates a fully calibrated package containing three-color RGB LED chips and drive IC, ensuring precise calibration before shipment.

Through a single microcontroller, up to 4079 devices can be daisy-chained and controlled. Moreover, the microcontroller can detect the temperature and fault status of each device.

The addition of CAN transceivers between daisy-chained devices ensures seamless long-distance data communication.

In an LED chain, each device is individually addressable via the serial bus. A proprietary bus protocol is used, optimized for this particular application.

Application

- Automotive: Backlight in Dashboard and Switch
- Automotive: Ambient lighting (Interior lighting)
- Functional lighting
- General lighting

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1. Features:

- AEC-Q100 Grade 2 Qualified
- AEC-Q102-003 Qualified
- Max 4KHz, 12-bit PWM Dimming resolution
- Temperature compensation for red channel
- 8-bit brightness resolution for red, green, and blue LED
- Maximum of 4079 devices in one LED chain
- Bidirectional, half-duplex, 2Mbps, CRC protected serial communication
- White point calibration D65
- Support for 16 multicast address groups
- On-die oscillator
- Built-in diagnostic functions
- Auto addressing of LED via the serial bus in daisy chain
- Supply voltage 4.5V ~ 5.5V
- ESD, HBM protection (AEC Q100-002 ANSI/ESDA/JEDEC JS-001): $\pm 2\text{kV}$
- Qualified according to JEDEC MSL 2

2. Packing outline Dimension

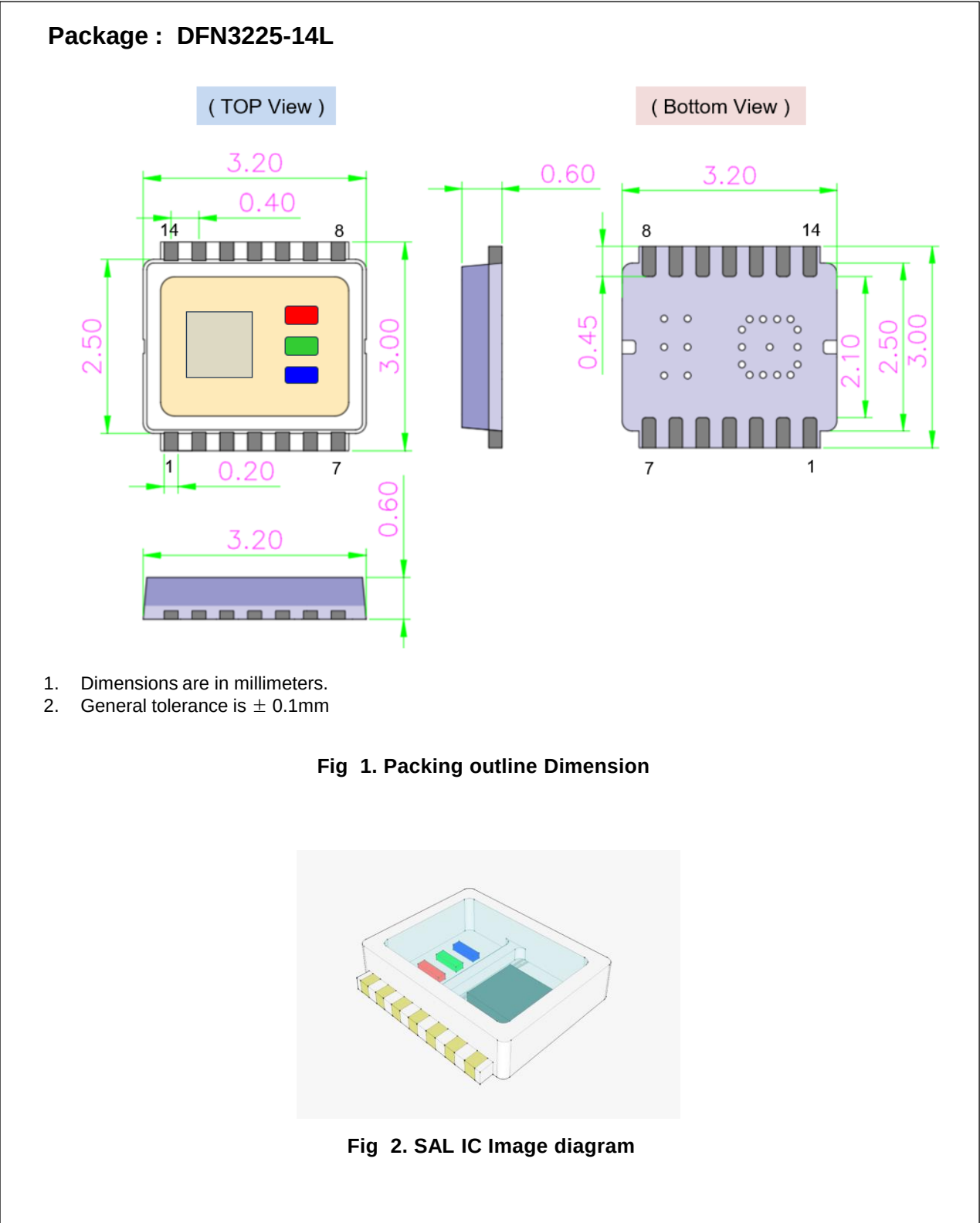


Fig 2. SAL IC Image diagram

3. Pin Configuration

DFN3225-14L

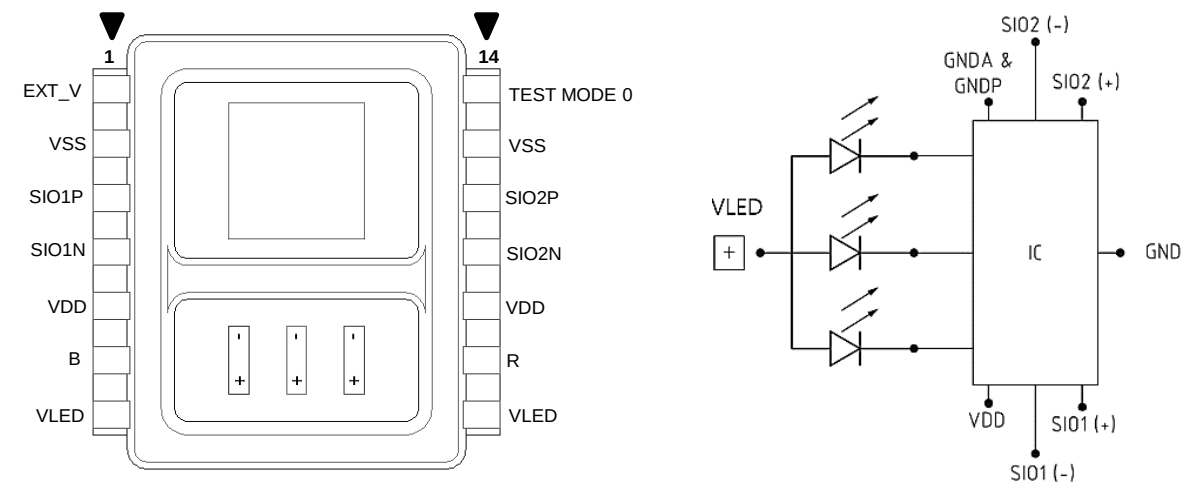


Fig 3. Top-View Pin Map and electrical internal circuit

Pad	Name	Description
1	EXT_V	ADC input for measuring of External analog voltage. If not in use, connect to ground.
2	VSS	Power Ground
3	SIO1_P	Serial communication interface master side, positive polarity
4	SIO1_N	Serial communication interface master side, negative polarity
5	VDD	5V DC Supply voltage for IC
6	N.C	No connection (BLUE LED Cathode side)
7	VLED	5V DC Supply voltage for RGB LEDs
8	VLED	5V DC Supply voltage for RGB LEDs
9	N.C	No connection (RED LED Cathode side)
10	VDD	5V DC Supply voltage for IC
11	SIO2_N	Serial communication interface slave side, negative polarity
12	SIO2_P	Serial communication interface slave side, positive polarity
13	VSS	Power Ground
14	TM0	Connect to GND (Test mode entry pin with a built-in pull-down resistor, 200KΩ)

Table 1. Pin Description

3. Equivalent Circuits of individual pins

Pad Name	I/O	Internal Circuit
V_{DD} , V_{LED}	-	
SIOx_N, SIOx_P, TMO	I/O	
EXT_V	I	
V_{SS}	-	

4. Absolute Maximum Ratings (unless otherwise noted) ⁽¹⁾

Parameter	Symbol	Rating	Unit
DC Supply voltage	V _{DD}	-0.3 ~ 7.0	V
LEDs Anode Supply voltage	V _{LED}	-0.3 ~ V _{DD} + 0.3	V
Test mode entry voltage	V _{TEST}	-0.3 ~ V _{DD} + 0.3	V
External analog voltage output	V _{EXT_V}	-0.3 ~ V _{DD} + 0.3	V
Input voltage range	SIO1_P/N	-0.3 ~ V _{DD} + 0.3	V
Output voltage range	SIO2_P/N	-0.3 ~ V _{DD} + 0.3	V
Junction to ambient thermal resistance ⁽²⁾	θ _{JA}	TBD	°C/W
Junction to case thermal resistance ⁽²⁾	θ _{JC}	TBD	°C/W
Soldering Temperature	Reflow	260°C 30 sec Max.	°C
Operating Temperature	T _{OPR}	-40 ~ +125	°C
Storage Temperature	T _{STG}	-40 ~ +125	°C
Junction Temperature	T _J	-40 ~ +125	°C
ESD Sensitivity	HBM ⁽³⁾	±2.0	kV
	CDM ⁽⁴⁾	±500	V
	CDM(coner pin)	±750	V

Table 2. Absolute Maximum Ratings

Note

⁽¹⁾ Stresses beyond those listed under Absolute Maximum Ratings may cause permanent damage to the device.

⁽²⁾ Power dissipation is limited by package type and the temperature of the operating environment.

⁽³⁾ AEC Q100-002 indicates that HBM stressing shall be in accordance with the ANSI/ESDA/JEDEC JS-001.

⁽⁴⁾ AEC Q100-011

5. Recommended Operating Condition (V_{DD} =5.0V, T_A =25.0°C unless otherwise noted)

Parameter	Symbol	Min.	Typ.	Max.	Unit
DC Supply voltage	V _{DD}	4.3	5	5.5	V
LEDs Anode Supply voltage	V _{LED}	4.3	5	5.5	V
Serial I/O Voltage	SIO1_P, SIO1_N	4.5	5	5.5	V
V _{DD} Ramp up time	T _{RAMP_VDD}	0.5			ms
Device ready for commands time	T _{INIT}			5.0	ms
Data rate for communication	f _{COMM}	1.8	2.0	2.2	Mbps
Programming voltage	V _{PP}		0	30	mV
Ambient temperature	T _A	-40		105	°C

Table 3. Recommended Operating Condition

Important statement

Global Technologies reserves the right to change the above-mentioned information without prior notice.

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6. Current Consumption (V_{DD} =5.0V, T_A =25.0°C unless otherwise noted)

Parameter	Symbol	Min.	Typ.	Max.	Unit
LED Red current	I _{R_AVERAGE}		13.3		mA
	I _{R_PEAK}				mA
LED Green current	I _{G_AVERAGE}		11.5		mA
	I _{G_PEAK}				mA
LED Blue current	I _{B_AVERAGE}		5.2		mA
	I _{B_PEAK}				mA
V _{DD} Deep sleep mode current	I _{Q_DPSLEEP}		TBD		uA
V _{DD} Sleep mode current (LED driver off)	I _{Q_SLEEP}		0.9		mA
V _{DD} Active mode current (LED driver on)	I _{Q_ACTIVE}	1.15 + (LEDs current/200)			mA

※ RGB set (255,255,255) **Table 4. Current Consumption**

7. Power on Reset (V_{DD} =5.0V, T_A =25.0°C unless otherwise noted)

Parameter	Symbol	Min.	Typ.	Max.	Unit
V _{DD} voltage for communication start	V _{POR}	-	-	3.8	V

Table 5. Power on Reset threshold voltage

8. Under voltage detection on V_{DD} (T_A =25.0°C unless otherwise noted)

Parameter	Symbol	Min.	Typ.	Max.	Unit
Undervoltage Lockout threshold	V _{UVD_RISING}	4.1	4.2	4.3	V
	V _{UVD_FALLING}	3.9	4.0	4.1	V
UVLO Detection hysteresis threshold	V _{UVD_HYST}		180		mV
UVLO Detection filter time	V _{UVD_FILTER}		62		us

Table 6. Under voltage detection voltage

9. Main Oscillator (T_A =25.0°C unless otherwise noted)

Parameter	Symbol	Min.	Typ.	Max.	Unit
Internal Main Oscillator frequency	F _{MCLK}	14.4 (-10%)	16.0	17.6 (+10%)	MHz

Table 7. Main Oscillator frequency clock

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10. Interface Characteristics (V_{DD} =5.0V, T_A=25.0°C unless otherwise noted)

Parameter		Symbol	Min.	Typ.	Max.	Unit
Allowed data rate for communication input		f _{COMM_DATA_INPUT}	1.8	2.0	2.2	MHz
Signal data rate for single ended Manchester code (Manchester coded signal = Real data rate x 2)		f _{ME_DATA_SINGLE}			4.4	Mbps
Allowed duty cycle variance for communication input		D _{COMM_DATA_INPUT}	45		55	%
Time between two messages (send from MCU)		t _{MSG_INTERVAL}		6 (12clk)		us
Input High Voltage Single ended		V _{IH_SE}	2.0			V
Input Low Voltage Single ended		V _{IL_SE}			0.8	V
Differential input amplitude		V _{IAMP_DIFF}	250	300	350	mV
Differential output amplitude		V _{OAMP_DIFF}	250	300	350	mV
Output driver pull-up resistance (I _{DRV_OUT} = -10mA)		R _{OUT_PU_SE}		15	TBD	Ω
Output driver pull-down resistance (I _{DRV_OUT} = -10mA)		R _{OUT_PD_SE}		15	TBD	Ω
Propagation delay from RX to TX on single-ended mode	Low to High	T _{DLY_RXTX_SING1}			50	ns
	High to Low	T _{DLY_RXTX_SING2}			50	ns

Table 8. Interface Characteristics

11. LVDS Physical layer parameters (V_{DD} =5.0V, T_A =25.0°C unless otherwise noted)

Parameter	Test Conditions	Symbol	Min.	Typ.	Max.	Unit
Common mode voltage with active communication		V _{LVDS_COMM}		1.2		V
Common mode voltage with no active communication	Guaranteed by design	V _{LVDS_OFF}		0		V
LVDS differential voltage	Guaranteed by design	V _{LVDS_DIFF}		300		mV
LVDS termination resistance	Included IC	R _{LVDS_TERM}	-10%	200	+10%	Ω
LVDS TX Current		I _{LVDS_TX}	-10%	1.5	+10%	mA
LVDS Pull down resistance		R _{LVDS_PD}		200		KΩ
Propagation delay from RX to TX on LVDS mode	Low to High	T _{DLY_RXTX_LVDS1}			50	ns
	High to Low	T _{DLY_RXTX_LVDS2}			50	ns

Table 9. LVDS Physical layer parameters

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12. RGB LED Driving current characteristic ($V_{DD}=5.0V$, $T_A=25.0^{\circ}C$ unless otherwise noted)

Parameter	Symbol	Min.	Typ.	Max.	Unit
Constant output current accuracy (All LED is on, $V_{OUTn}=1V$, $IDAC=max$)	$I_{O_LED_ACC}$	-5%	20	+5%	mA
Constant output current aging drift	$I_{O_LED_DRIFT}$	-1%		+1%	%
Constant output current headroom voltage ($IDAC=max$)	$V_{O_LED_HDRM}$	1		0.7	V
Load regulation of constant-current output (All LED is on, $V_{OUTn}=1V\sim 3V$, $IDAC=max$)	$I_{O_LED_ACC}$	-1%		+1%	%/V
Output leakage current (All LED is off, $V_{OUTn}=5V$)	$I_{O_LED_LEAK}$		0.1	1	uA

Table 10. LED Driving Current temperature Drift

13. RGB LED Driving current temperature Drift ($^*\Delta I_i = \text{Abs}((I_{25^{\circ}C@2V} - I_{temp}) / I_{25^{\circ}C@2V})$)

Parameter	Symbol	Test Conditions	Min.	Typ.	Max.	Unit
RED Driving current temperature drift	ΔI_{RED}^*	$T_A = -40^{\circ}C$, Max. current			5	%
		$T_A = 105^{\circ}C$, Max. current			5	
GREEN Driving current temperature drift	ΔI_{GREEN}^*	$T_A = -40^{\circ}C$, Max. current			5	%
		$T_A = 105^{\circ}C$, Max. current			5	
BLUE Driving current temperature drift	ΔI_{BLUE}^*	$T_A = -40^{\circ}C$, Max. current			5	%
		$T_A = 105^{\circ}C$, Max. current			5	

Table 11. RGB LED Driving current temperature Drift

14. Dimming parameter timing Characteristics ($T_A=25.0^{\circ}C$ unless otherwise noted)

Parameter	Symbol	Test Conditions	Min.	Typ.	Max.	Unit
LED current Rising time	$T_{SR_RS_LED}$	I_{OUTn} (10% to 90%)		0.3	1.0	us
LED current Falling time	$T_{SR_FALL_LED}$	I_{OUTn} (90% to 10%)		0.3	1.0	us
LED On command Propagation delay time	$T_{DLY_ON_LED}$	Command to 10% I_{OUTn}		30	50	ns
LED Off command Propagation delay time	$T_{DLY_OFF_LED}$	Command to 90% I_{OUTn}		30	50	ns

Table 12. Dimming parameter timing Characteristics

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15. Dimming Phase shift for RGB LED driving (V_{DD}=5.0V, T_A=25.0°C unless otherwise noted)

Parameter	Symbol	Test Conditions	Min.	Typ.	Max.	Unit
Red Phase shift refer to PWM dimming frequency	P _{SHIFT_RED}	Guaranteed by design		90		degree
Green Phase shift refer to PWM dimming frequency	P _{SHIFT_GREEN}	Guaranteed by design		0		degree
Blue Phase shift refer to PWM dimming frequency	P _{SHIFT_BLUE}	Guaranteed by design		270		degree

Table 13. Dimming Phase shift for RGB LED driving

16. LED Fault diagnostic (V_{DD}=5.0V, T_A=25.0°C unless otherwise noted)

Parameter	Symbol	Test Conditions	Min.	Typ.	Max.	Unit
LED Open diagnosis threshold voltage	V _{TH_OPEN}			250		mV
LED Open hysteresis threshold voltage	V _{TH_HYS_OPEN}			30		mV
LED Short diagnosis threshold voltage	V _{TH_SHORT}	SH_LVL[1:0] = 2b'10		4.25		V
LED fault diagnostic deglitch filter time	T _{DIAG_FILTER}	Guaranteed by design		16		cnt

Table 14. LED Fault diagnostic

17. Electro-optical Characteristics (V_{DD} =5.0V, T_A =25.0°C unless otherwise noted)

Parameter	Symbol	Color	Code	Conditions	Min.	Typ.	Max.	Unit
Dominant Wavelength ⁽¹⁾	λ _{D_R}	Red	W36	(255, 0, 0)	620	625	630	nm
	λ _{D_G}	Green	W25	(0, 255, 0)	520	530	540	nm
	λ _{D_B}	Blue	W12	(0, 0, 255)	450	465	480	nm
Luminous Intensity ⁽²⁾	I _W	White ⁽⁴⁾		(255,255,255)	-	3000	-	mcd
	I _R	Red	L634	(255, 0, 0)	1000	-	1100	mcd
Radiant Flux	I _G	Green	L060	(0, 255, 0)	11.0	-	12.5	mW
	I _B	Blue	L062	(0, 0, 255)	14.0	-	15.5	mW
Forward Voltage	V _{F_R}	Red	D	@10mA	2.0	-	2.1	V
	V _{F_G}	Green	K	@10mA	2.6	-	2.7	V
	V _{F_B}	Blue	L	@10mA	2.8	-	2.9	V
View angle ⁽³⁾	φ	-		(255,255,255)		120		degree

Table 15. Electro-optical Characteristics

- Note (1) Dominant wavelength has an accuracy of ±2nm
(2) Luminous Intensity is tested by a tester calibrated by CAS 140B(CIE LED_B) and has an accuracy of 10%
(3) Viewing angle is the angle until 50% of brightness measured from the front part of LED.
(4) D65(6500K), The above luminous intensity represent brightness at 100%.

18. Analog Digital Converter (TBD) ($V_{DD} = 5.0V$, $T_A = 25.0^{\circ}C$ unless otherwise noted)

Parameter	Test Conditions	Symbol	Min.	Typ.	Max.	Unit
Bit Resolution		V_{TH_OPEN}		10		bit
Sampling Rate		F_{SMP_RATE}		250		KSPS
Signal to Noise Ratio		SNR	52	55		dBc
Signal to Noise and Distortion Ratio		SNDR	50	54		dBc
Spurious Free Dynamic Range		SFDR	60	62		dBc
ADC Value for EXT_V	EXT_V=1.0V, Register read	ADC_{EXT1V}		TBD		digit
ADC Value for EXT_V	EXT_V=2.5V, Register read	ADC_{EXT2V}		TBD		digit
ADC Value for EXT_V	EXT_V=4.0V, Register read	ADC_{EXT3V}		TBD		digit
ADC Value for Temp Sen.	$T_A = 25^{\circ}C$, Register read	ADC_{Troom}		0x200		digit
ADC Value for Temp Sen.	$T_A = 105^{\circ}C$, Register read	ADC_{Thot}		TBD		digit
Integral Nonlinearity		INL	-2		+2	LSB
Differential Nonlinearity		DNL	-1		+1	LSB
Offset error		ERR_{OFFSET}	-4		+4	LSB
Input Range for External	EXT pin	V_{EXT_INPUT}	0.5		4.5	V
Input range for Temp Sen.		V_{TEMP_INPUT}	0.75		1.65	V

Table 16. Analog Digital Converter

19. Lookup Table for Red LED Non-Linear Temperature Compensation

LUT	Values
TEMP_LUT_TC1	000
TEMP_LUT_TC2	000
TEMP_LUT_TC3	000
TEMP_LUT_TC4	000
TEMP_LUT_TC5	000
TEMP_LUT_TC6	000
TEMP_LUT_TC7	000
TEMP_LUT_TC8	000
TEMP_LUT_TC9	000
TEMP_LUT_TC10	000

Table 17. Lookup Table for Red LED

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20. Color Coordinates Group

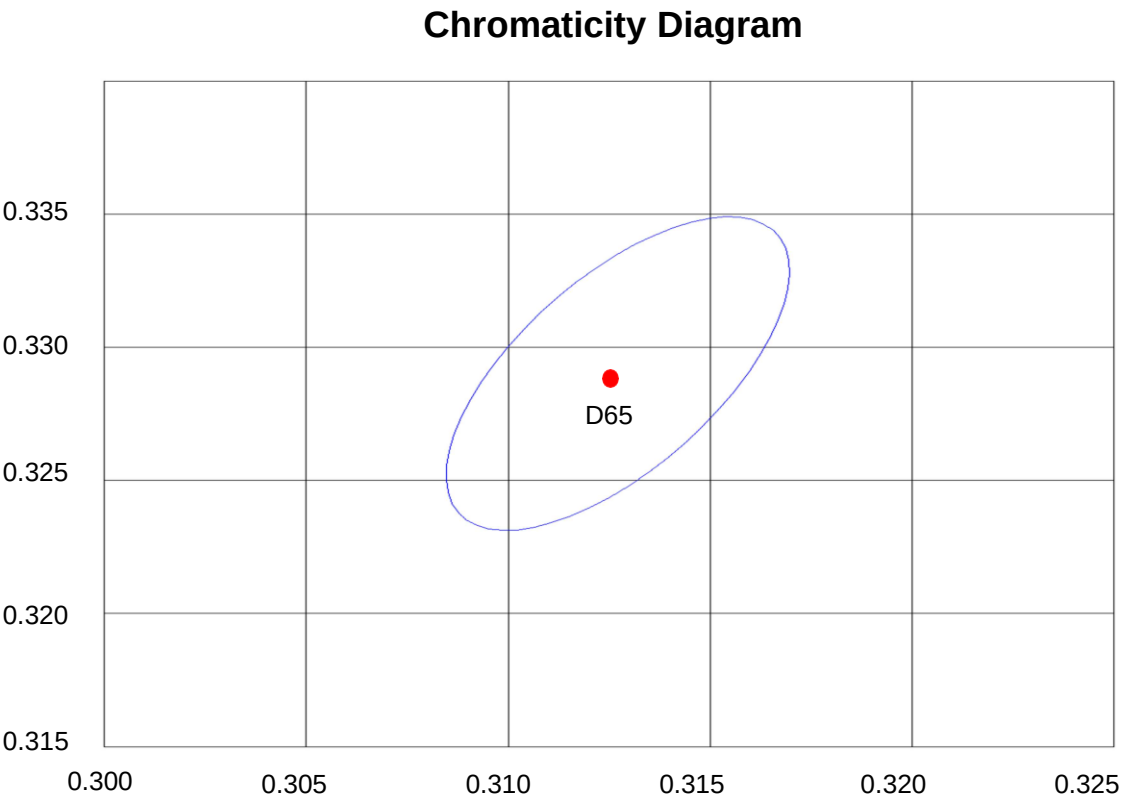


Fig 4. Chromaticity Diagram

Rank	Description	Center		Radius		Angle(°) θ
		x	y	a	b	
D65	Macadams 3steps	0.320	0.331	0.00669	0.00285	58.57

Table 18. Color Coordinates Group

21. Block Diagram

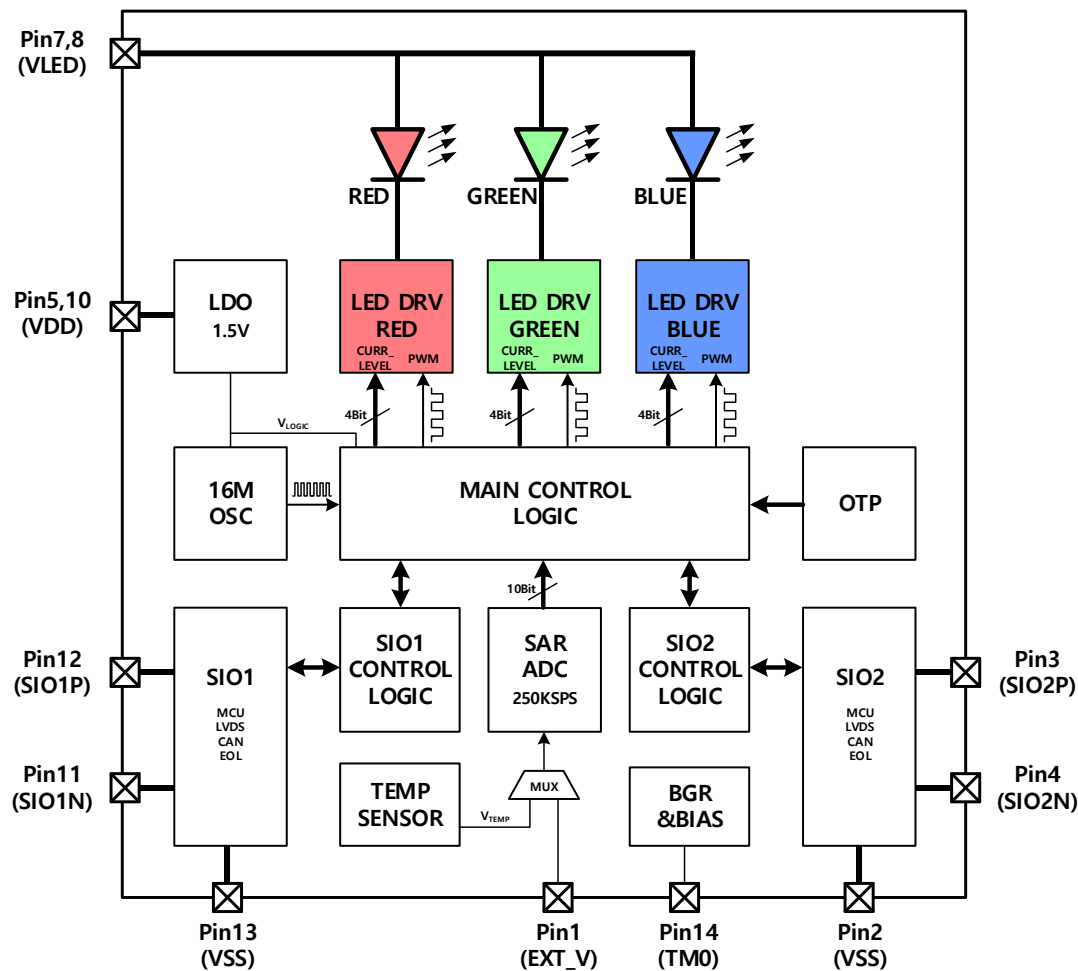


Fig 5. Block Diagram

The device implements a communication for the reception of control commands and for providing device status and configuration data. Low side, configurable constant current sinks are provided for controlling 3 LEDs (i.g. RGB). The main control logic computes the PWM duty cycles from the incoming commands and applies the corresponding control values to the three PWM generators. The main control logic is also in charge of a periodic temperature monitoring and an appropriate duty cycle adjustment for the red PWM channel. The actual device temperature is obtained via an integrated analog-digital converter (ADC). Besides the temperature the ADC can also measure an external voltage through the EXT_V PIN. The die temperature and external voltage measurements are periodically updated in the corresponding registers, and the values can be read by the MCU by host commands. As each device is individually calibrated to compensate for LED variations, the corresponding parameters can be stored in an one-time-programmable memory (OTP). The calibration data stored in the OTP is loaded into the main control logic whenever the IC is turned on or reset command.

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22. Typical Application Layout

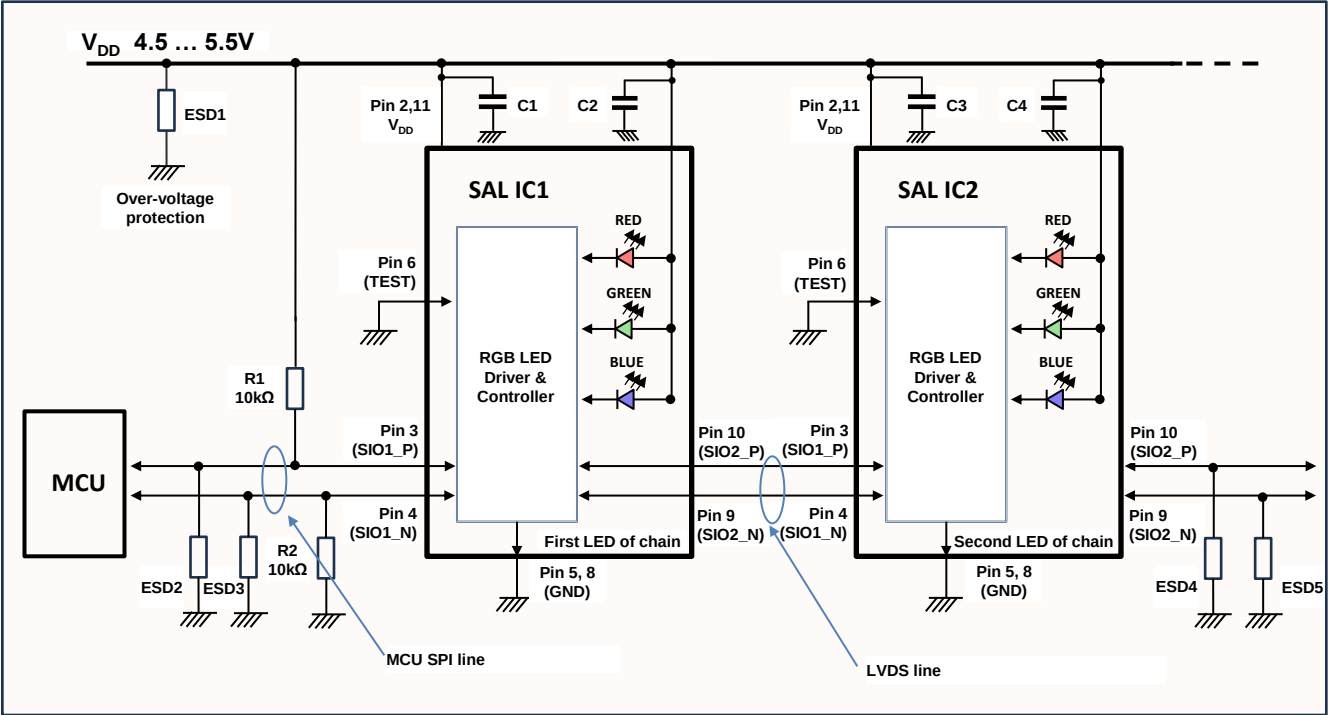


Fig 6. Typical Application Layout

In order to prevent voltage drops, it is recommended to mount the capacitors C1 closely to the V_{DD} pin. The dimensioning of the capacitors depends on the PCB layout and the supply concept.

23. Thermal Considerations

Need to suggest the maximum amount of power that the IC can consume under AEC-Q100 grade 2 conditions(T_A 105°C at still air). Thermal resistance characteristics would be variable depending on the IC, package, and PCB layer characteristics.

“Example of allowable IC power dissipation characteristics by ambient temperature”

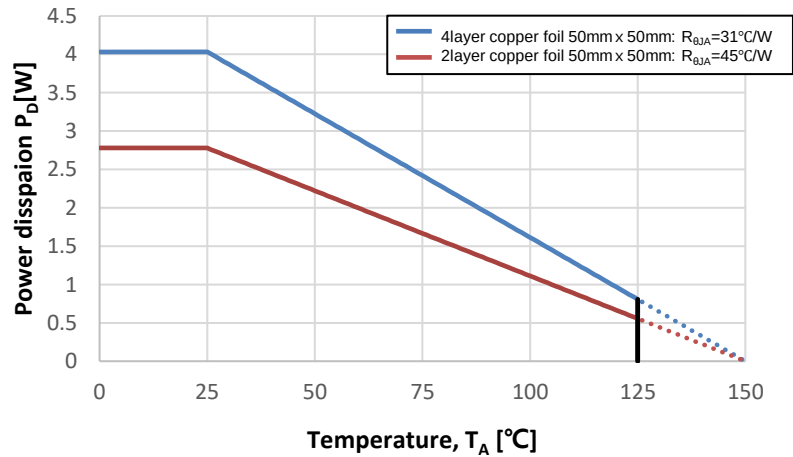


Fig 7. Thermal Considerations

24. Functional Description

24-1. Device States

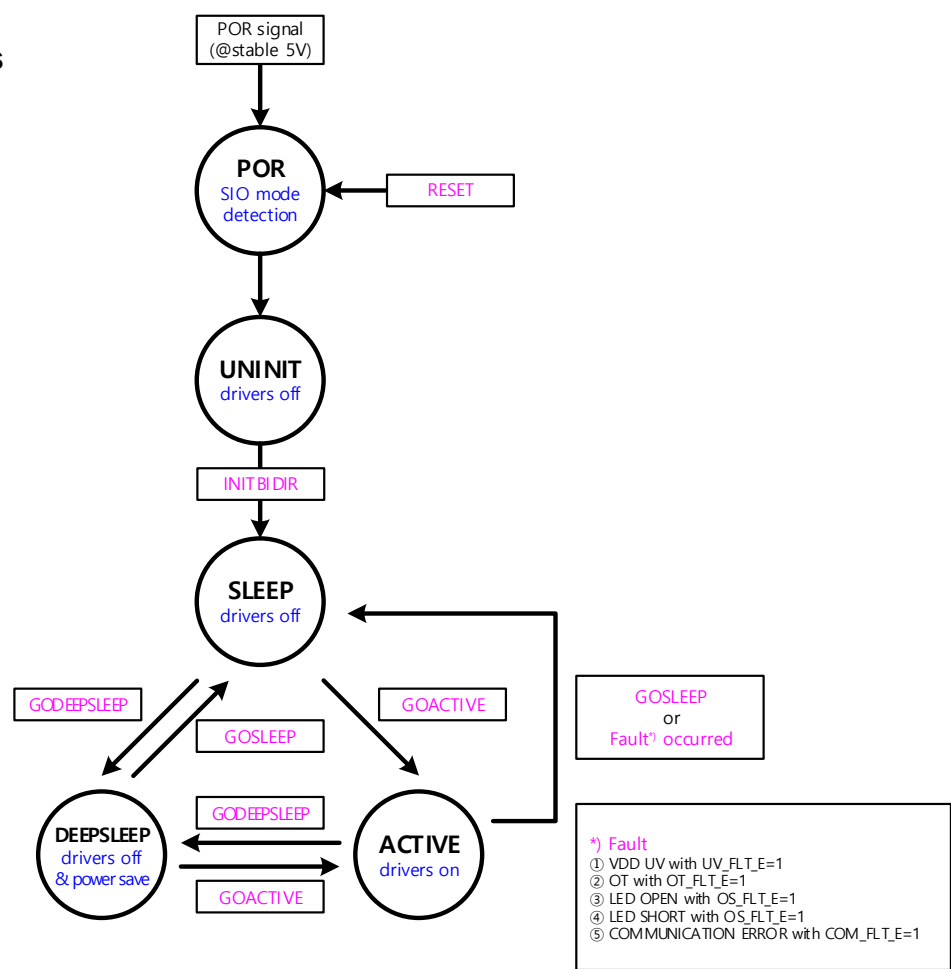


Fig 8. IC State

POR : The ID804 is in an off state.
UNINIT : The ID804 will reach this state automatically after power-up. PWM values are 0 and LED drivers disabled.

In this state, the ID804 only responds to a limited set of commands:

- INITBIDIR
- RESET (broadcast)
- CLRERROR (broadcast)
- SET_SETUP1 (broadcast)
- SET_SETUP2 (broadcast)
- SET_TEMPTH (broadcast)
- SET_TEMPHYS (broadcast)
- SET_RGB (broadcast)

Other telegrams are ignored and fault flag (STATUS2.COM_FLT) set. Broadcast telegrams are fast forwarded and addressed telegrams are forwarded after the telegram has been fully received.

SLEEP : The ID804 enters SLEEP state after receiving an INITBIDIR command, after a failure event (if selected) or via GOSLEEP command.

In this state, the LED drivers are disabled but the last PWM parameters are stored, i.e., not reset to 0. An update of the PWM parameters is possible.

ACTIVE : The ID804 enters this mode after receiving the GOACTIVE command if no fault conditions are present.

In this state the LED drivers are enabled with the respective current setting, PWM frequency and duty cycle for each channel. An update of the PWM parameters is possible.

After entering the ACTIVE state and LED drivers enabled, the open/short detection is executed once for every channel with a PWM value exceeding the minimum value. See Chapter 25-5.

DEEPSLEEP : The ID804 is in a power save mode with reduced current consumption. In this state, the ADC is disabled, and the LVDS receiver waits in a low power mode. When a signal is detected on the LVDS, the receiver switches to normal mode, transmits the data to the internal logic, and then returns to low power mode until the next LVDS signal is received.

24-2. Device Enable

When the supply voltage is connected and VDD increases above the V_{POR} level, or the RESET command is received, the ID804 undergoes the following startup sequence (Fig 11. Startup behavior of the ID804. The example shows the MCU and EOL modes. After a RESET command, the device starts in the POR state.)

- 1. All registers are initialized to their default value (POR in Fig 11).
- 2. Trim settings are loaded and verified. A mismatch sets the OTPCRC flag in STATUS2.
- 3. Voltage levels at SIOx pins detect communication modes (shaded signals in Fig 11).
- 4. The ID804 enters the UNINIT mode and is now ready to receive the first command (UNINIT in Fig 11).

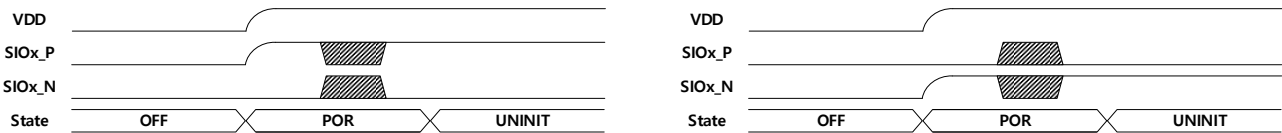


Fig 9. Startup behavior of the ID804. The example shows the MCU (left) and EOL (right) modes. After a RESET command, the device starts in the PoR state.

Note : The whole sequence takes max. 5 ms.
(When sequence is started caused by RESET command, the The whole sequence takes approximately 1 ms.)

24-3. Communication

1. Single Ended, LVDS Link and Data Structure

(1) Overview

- Bi-directional
- LVDS - differential Manchester coded
- Single ended – Manchester coded & Manchester decoded or NRZ(clock & data) Push-pull interface, clock is invertible
- Communication direction and SIOx connection is flexible
- Communication speed is typ. 2Mbps Manchester coded

(2) Communication modes

1. Single Ended, LVDS Link and Data Structure

Each one of the 2 available SIOx interfaces do support 4 different modes.

Each SIOx only can operate in semi-duplex mode meaning it either does receive or transmit a telegram at the same time. However, it is possible that a device does receive a telegram at one SIOx and start transmitting at the other SIOx interface.

The 4 communication modes are:

1.“LVDS-mode”:

The SIOx communication is using LVDS signaling for transmitting and receiving a communication telegram. The bit stream is using Manchester encoding to transfer DATA & CLK on the same line.

2.“MCU-mode”: In this mode a SIOx is using single-ended mode for the signaling. Received telegrams are Manchester encoded on SIOx_P port. For transmitting, the device is decoding the Manchester signal. SIOx_P will deliver the DATA in NRZ format and SIOx_N delivers the corresponding clock.

3.“EOL(End Of Line)-mode”: SIOx port does consider itself to be at the end of the daisy chain.

4.“CAN-mode”: In this mode, it is possible to communicate via a CAN FD transceiver. SIOx_P is always in output state and transmits telegram as Manchester encoded signals (single-ended), SIOx_N is always in input state and receives telegrams as Manchester encoded signal (single-ended).

24-3. Communication

2. Communication mode selection

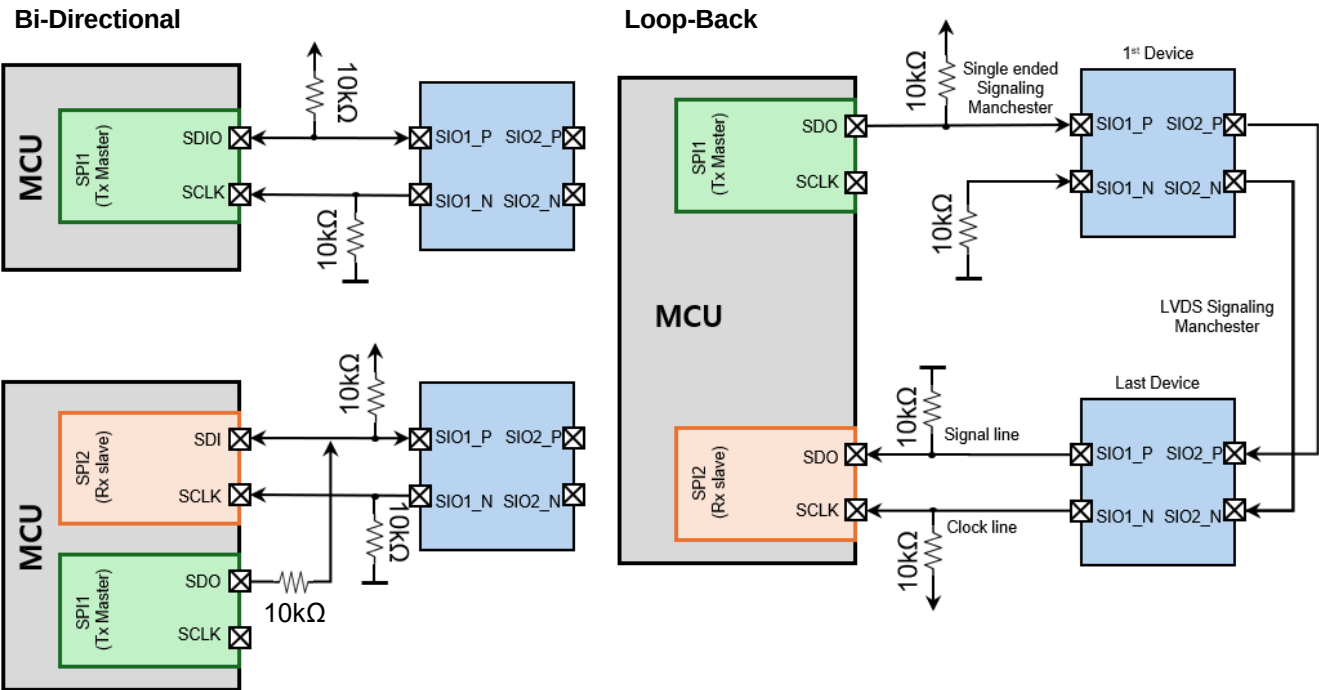
Communication mode auto detection takes place during initialization phase via voltage level check on SIOx_P & SIOx_N and this mode cannot be changed during the operation cycle. After a HW Reset (POR) or RESET command received the Initialization with mode selection take place again.

Mode	R_Mode ¹⁾ SIOx_P	R_Mode ¹⁾ SIOx_N	Input		Output	
			SIOx_P	SIOx_N	SIOx_P	SIOx_N
MCU	Pull-Up	Pull-Down	ME	-	Data	CLK
CAN	Pull-Up	Pull-Up	-	ME	ME	-
LVDS	None	None	ME	ME	ME	ME
EOL ²⁾	Pull-Down	Pull-Up	-	-	Data	CLK

Note:
1. Pull-Up & Pull-Down resistors Typical Value : 10kΩ
2. EOL : End of Line

Fig 10. Communication mode selection Table

3. MCU-mode communication physical layer and bit coding



Single ended communication in the master to slave direction is performed with so-called Manchester encoded. In that case a bit stream is sent via SIOx_P only; SIOx_N of the device is ignored. Slave to master communication is performed with NRZ coding. In that case SIOx_N provides a clock signal. The clock signal is inverted if it's selected via setup register. Communication Data rate effective is typ. 2Mbps. All single-ended connections should be kept as short as possible to ensure signal integrity and to lower EMC concerns.

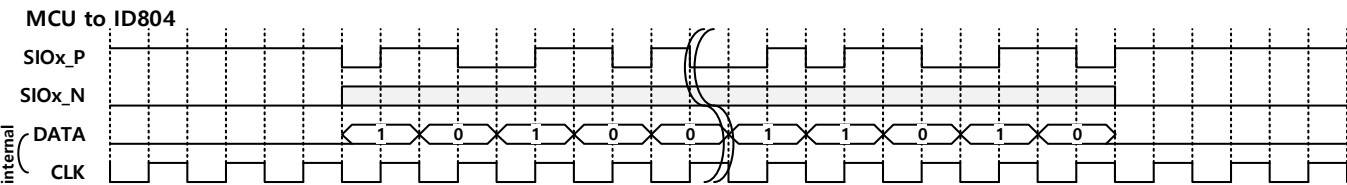


Fig 11. Example of MCU to ID804 communication

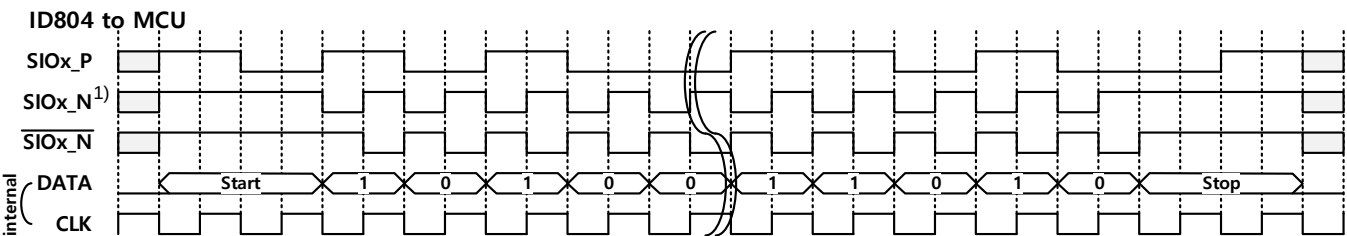


Fig 12. Example of ID804 to MCU communication

Note:
1. The receiving clock could be inverted via setup register (SETUP2.CLK_INV_E)

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24-3. Communication

4. LVDS-mode communication physical layer and bit coding

For differential communication the termination resistance R_{LVDS_term} is 200 Ohm, differential voltage V_{LVDS_diff} is typ. 300 mV, data rate is typ. 2 Mbps. Common mode voltage V_{LVDS_comm} is typ.1.2 V, it means the max voltage on the communication lines during communication is typ. 1.35 V and the min. voltage is typ. 1.05 V

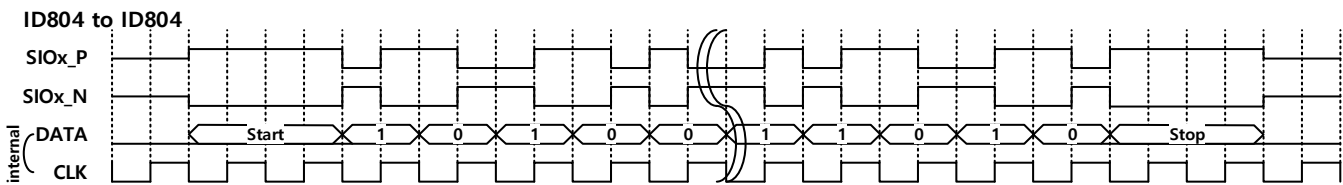


Fig 13. Example of ID804 to ID804 communication

5. CAN-mode communication physical layer and bit coding

The addition of Standard CAN-FD transceivers between daisy-chained ID804 devices in a point-to-point fashion ensures seamless long-distance data communication.

The mode is selected by pull-up resistors on the SIOx_P and SIOx_N pins. SIOx_P transmits Manchester-encoded signals to the CAN-FD transceiver, while SIOx_N receives them. During transmission, any signals received on SIOx_N are ignored. CAN-FD transceivers are required for voltage rating compliance, ID804 can not the support a bus topology of CAN-FD protocol.

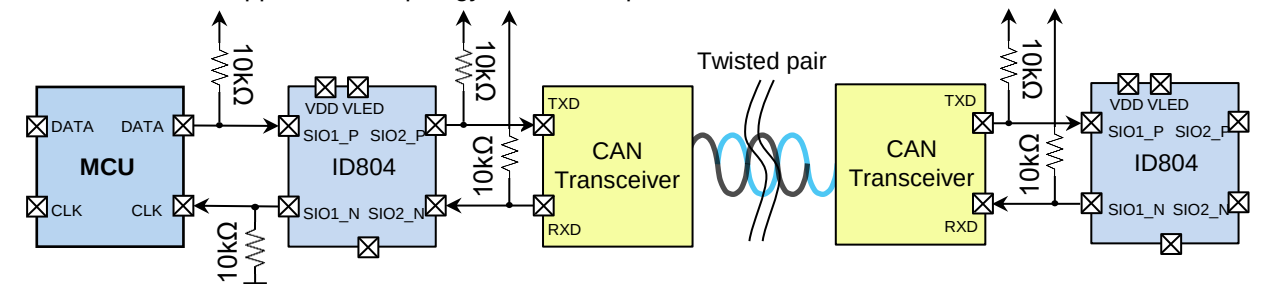


Fig 14. Bi-directional chain example with CAN-FD connection. The ID804 serves as Manchester decoder between the chain and the MCU(upstream direction).

The MCU can be programmed for direct communication but needs a Manchester decoder.

Additional components may be required based on the transceiver type.

All single-ended connections should be kept as short as possible to ensure signal integrity and to lower EMC concerns.

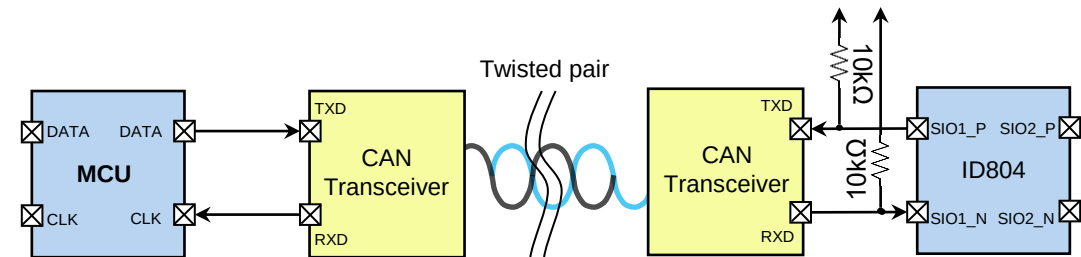


Fig 15. Bi-directional chain example with direct MCU-CAN-FD connection. The MCU needs to implement the Manchester decoder.

If a transceiver is detected on SIOx the output and input message is via single ended Manchester encoded communication with typ. 2 Mbps data transfer rate. For the transceiver the data transfer rate depends on the data signal due to the Manchester encoded data. Means up to the double transfer rate is needed (4 Mbps). Therefore, the use of a 5Mbps CAN-FD transceiver is required.

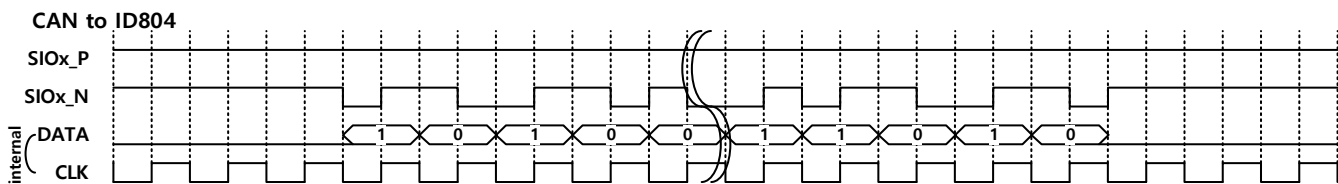


Fig 16. Example of CAN Transceiver to ID804 communication

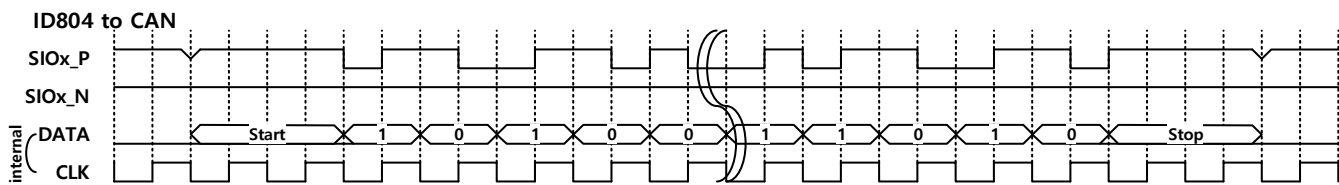


Fig 17. Example of ID804 to CAN Transceiver communication

24-3. Communication

6. Message Frame Format

ID804 has 3 types of frame format according to DATA bit. The format is the same for upstream and downstream communication in case of data bit is 12 or 24. 0bit data frame format Fig 21.

0bit Data Frame Format Fig 21 can use only downstream.

Byte 1								Byte 2								Byte 3								Byte 4								Byte 5								Byte 6								Byte 7													
55	54	53	52	51	50	49	48	47	46	45	44	43	42	41	40	39	38	37	36	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0						
PREAMBLE								DEV_ADDR [11:0]								COMMAND [7:0]								DATA [23:0]																														CRC [7:0]							

Fig 18. 24bit Data Frame Format

Byte 1								Byte 2								Byte 3								Byte 4								Byte 5								Byte 6							
47	46	45	44	43	42	41	40	39	38	37	36	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PREAMBLE				DEV_ADDR [11:0]								COMMAND [7:0]								DATA [11:0]								DUMMY [3:0]				CRC [7:0]															

Fig 19. 12bit Data Frame Format

Byte 1								Byte 2								Byte 3								Byte 4							
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PREAMBLE				DEV_ADDR [11:0]								COMMAND [7:0]				CRC [7:0]															

Fig 20. 0bit Data Frame Format

- PREAMBLE (4 bit) : Fixed to 4'b1010.
Avoid messages with incorrect preamble as this could lead to forwarding of garbage telegrams along the chain and to raising the STATUS2.COM_FLT.
- DEV_ADDR (12 bit) : Allowed values for downstream messages are 0x000 (broadcast) and 0x001 to 0xFEE (unicast nodes) and 0xFEf to 0xFFE (multicast). Upstream messages always contain the device address of the sender in the DEV_ADDR field. Not all commands are allowed in a broadcast message. See Chapter 25-6.
- COMMAND (8 bit) : See Chapter 25-6
- DATA (24bit or 12bit or 0bit) : DATA field has bits according to COMMAND. See Chapter 25-6
- DUMMY (4bit) : In case of 12bit Data Frame Format, 4bit DUMMY data (4'b1010) inserted to match the data size to BYTE.
- CRC (8 bit) : See Chapter 25-3.7.(8)

24-3. Communication

7. Message Handling

(1) Broadcast

An address of 0x000 initiates a broadcast, i.e., the same message is sent and received by all nodes on the chain. Whether a command can be broadcast is detailed in Chapter 25-6.

Broadcast Read or SR command is first received by all devices via the frame in downstream direction. The last node in the chain then immediately transmits its response frame upstream. The response frame's data field depends on the actual command. The response frame's address field is set according to the own device's address. All the nodes upstream forward all received response frames until a frame with the address of their adjacent node is received. Then the respective node transmits its own response frame. This procedure lasts until the chain's first node has transmitted its response frame.



Fig 21. Example of broadcast write

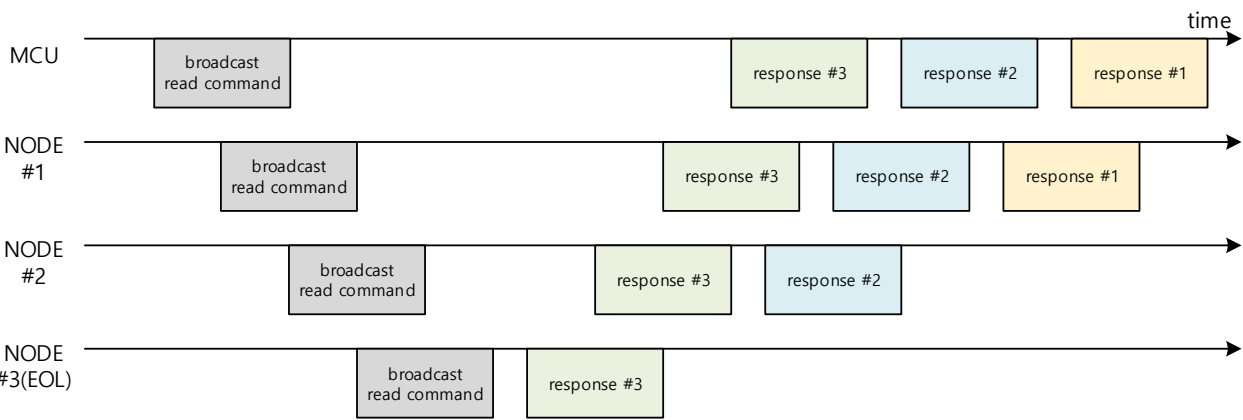


Fig 22. Example of broadcast read(or SR)

(2) Multicast

An address of 0xFEf to 0xFFE initiates a multicast, i.e., the same message is sent and received by same defined group nodes on the chain. Multicast group can be defined by MCAST command. Whether a command can be multicast is detailed in Chapter 25-6.

24-3. Communication

7. Message Handling

(3) Forwarding and Fast Forwarding

Message forwarding refers to the direct forwarding of a message without modification.

Messages are forwarded to the next device if the address in the message frame is the broadcast address, or multicast address, or it is different from the node's address.

If a message is to be forwarded, forwarding starts exactly one cycle after the preamble and address field are fully received, i.e., after 17 bits. This is referred to as fast-forwarding.

When a node is uninitialized, broadcast messages are fast-forwarded and multicast(0xFE to 0xFF) & unicast(0x001 to 0xFEE) & abnormal(0xFFF) addressed messages are forwarded only after the full telegram has been received.

(4) Responses and SR (Status Requests)

Some commands trigger a response from the addressed node. In this case, the address field contains the address of the sending node to indicate the origin of the message. Once the node receives the end of the command frame, the corresponding response will occur approximately 3 μ s (in case of broadcast) or 27 μ s (in case of unicast) later.

Some commands do not send a response but allow a SR(Status Request). The SR(Status Request) is done by specific command. See Chapter 25-6. In case of broadcast response or SR see Chapter (1).

If a valid SR(Status Request) command has been received, the node responds by sending the TEMP & STATUS2 registers to the MCU after the original command has been executed. Status bits are cleared after responding to the SR(Status Request). In the case of the CLRERROR command, the status bits are cleared by the command before the status request is sent.

Commands supporting a status request are marked in the command description. See Chapter 25-6.

Note that the ID804 does not allow simultaneous upstream and downstream telegrams on the same node, as this could lead to dropped or corrupted messages.

(5) Waiting Time between Telegrams

ID804 is not accepting a new telegram while a telegram is being received or transmitted on the other port. Likewise, simultaneous upstream and downstream messages are not supported.

Therefore, a waiting time has to be added by the MCU between sending two telegrams in order to ensure that all receiving and transmitting activities have been concluded.

The minimum waiting time between two downstream telegrams should be 6 μ s (edge to edge) and applies to write-only or without response request commands. If an answer has been requested, the next telegram should only be sent after the response has been received by the MCU.

24-3. Communication

7. Message Handling

(6) Communication Errors

The ID804 requires certain communication errors to be detected by each unit in a daisy chain. The below Fig 24 summarizes all relevant error conditions.

IC State	Preamble	Address	Command	CRC ^(*)	Non Target DEV		Target DEV	
					COM_FLT flag	Data Forwarding	COM_FLT flag	Data Forwarding
Uninitialized state (UNINIT)	Incorrect	-	-	-	Set	Stop	Set	Stop
	Correct	Unicast	-	-	Set	No stop	-	-
		Multicast	-	-	Set	No stop	-	-
		Broadcast	Incorrect	-	-	-	Set	No stop
			Correct	Incorrect	-	-	Set	No stop
Initialized State (SLEEP, NORMAL, DEEPSLEEP)	Incorrect	-	-	-	Set	Stop	Set	Stop
	Correct	Unicast	Incorrect	-	No set	No stop	Set	Stop
			Correct	Incorrect	No set	No stop	Set	Stop
		Multicast	Incorrect	-	No set	No stop	Set	No stop
			Correct	Incorrect	No set	No stop	Set	No stop
		Broadcast	Incorrect	-	-	-	Set	No stop
			Correct	Incorrect	-	-	Set	No stop

Fig 23. Communication Errors Table

Note :
(1) applies if CRC check is enabled.

The ID804 performs some error detection even for messages that are to be forwarded(CRC). If an error is detected in such a case, the COM_FLT flag is set and the message is still forwarded, i.e., communication is not stopped.

In UNINIT state(Uninitialized state), only broadcast address(0x000) and limit set of commands can accept "Correct" message. Other address (0x001 ~ 0xFFFF) messages are ignored. And still forwarded and set COM_FLT flag.

Errors due to incorrect command is only detected if the unit is addressed by unicast or multicast or broadcast.

24-3. Communication

7. Message Handling

(7) Communication Timeout

The READ commands or SR WRITE commands trigger a response from the addressed or all nodes. Each node needs to await all responses originating from the nodes downstream. Thereafter either the node's own response is transmitted or new commands are accepted. If new command(downstream) is received during response waiting, it is ignored and COM_FLT flag set on received node. Only the last node in the LED chain may immediately transmit its response. In case there is an error with the chain, not all expected responses may arrive. Thus, each of the commands expecting a response waits for a certain time only and then returns to its previous state without having transmitted the node's response data. The lengths of the timeouts depend on register setting command (SET_TIMEOUT). They are calculated to account for the worst-case oscillator frequency tolerance. I.e. the waiting node has a high-speed clock and all the nodes waited for have a low speed clock. The hardware implementation uses an internal clock for the timeout counter. The resolution is internal clock(Fosc_main(Typ. 16MHz)) * (2¹⁰).

Command	Counter Value	Min. timeout	Typ. timeout	Max. timeout	Units
SR WRITE (Downstream DATA field : 0bit)	TBD	TBD	TBD	TBD	ms
SR WRITE (Downstream DATA field : 12bits)	TBD	TBD	TBD	TBD	ms
SR WRITE (Downstream DATA field : 24bits)	TBD	TBD	TBD	TBD	ms
READ (Upstream DATA field : 12bits)	TBD	TBD	TBD	TBD	ms
READ (Upstream DATA field : 12bits)	TBD	TBD	TBD	TBD	ms

Fig 24. Timeout Specification

(8) Cyclic Redundancy Check

The CRC field contains the CRC8 check word calculated over the full message excluding the CRC field.

CRC Implementation: $x^6 + x^5 + x^4 + x^3 + x^2 + x + 1$.

Polynomial: 0x2F (normal notation)

Initial value: 0x00

XOR value: 0x00

Direction: MSB first

Note that this implementation uses the same polynomial but is not compatible with the AUTOSAR implementation because of the different initial and XOR values of 0xFF.

Note: The calculation direction for the CRC field in the non-volatile memory is LSB first.

24-4. Color Control and Temperature Compensation

1. PWM Generation

The ID804 incorporates three independent PWM channels, one for each LED. The resolution is 12bit. The supported duty cycles are 0/4096 to 4095/4096. The PWM output frequency is set by SETUP1.F_PWM_DIV register setting. The frequency is decided by (Fosc_main(Typ. 16MHz)) / (F_PWM_DIV + 1). The PWM frequency is reduced to the quarter of set frequency when RGB data is less than decimal 8. The output frequency is determined independently for each of the PWM channels.

(1) PWM Update

When a new PWM duty cycle has to be applied, this is always done at the end of a PWM cycle. I.e. the PWM always completes an output cycle using the previously active duty cycle and starts the next output cycle using the updated duty cycle.

(2) PWM Shift

In order to spread the current consumption of the LEDs over time, a phase shift enabled between the three PWM channels. It retains even if the output frequency of the channels is different. If a channel is operating at a lower frequency, it may be considered to leave out one or three full PWM cycles. When leaving DEEPSLEEP state, the channels are restarted appropriately to again obtain the correct phase shift. The fixed phase shift is defined in the following Fig 26. Please note the absolute phase shift times are nominal values. I.e. they are subject to vary with the internal oscillator's frequency.

PWM Channel	Related Phase Shift	Unit
Green	0	Degree
Red	90	Degree
Blue	270	Degree

Fig 25. PWM Phase Shift

2. Brightness Control Scaler

In operation, the color brightness of each LED is controlled via an 8-bit RGB register. This provides a color space of 3 x 8 bit = 24 bit. The 8-bit RGB value is scaled up to the 12-bit value using by Brightness calibration value(PWM max value). The PWM max values is stored OTP. The scaling equations of the red and green and blue channel are given in Fig 27.

Color	Scaling equation
Red	$PWM_{RED} = (PWM\ max \times R\ data + 128) \gg 8$ (When SETUP1.TC_E = 0, another case see Fig 27)
Green	$PWM_{GREEN} = (PWM\ max \times G\ data + 128) \gg 8$
Blue	$PWM_{BLUE} = (PWM\ max \times B\ data + 128) \gg 8$

Fig 26. Scaling equation

24-4. Color Control and Temperature Compensation

3. Temperature Monitor

For temperature detection, the ID804 provides an analog temperature sensor. This can be measured by ADC and value can be read from TEMP.TEMPERATURE register through READ_TEMP command.

Parameter	Description	Equation	Unit
T _{J,det}	Detected junction temperature	$T_{j,mon} = 25 + \frac{(TEMPERATURE - 512)}{3.6}$	°C
TEMPERATURE[9:0]	Detected Temperature value of ADC	-	digit

Fig 27. Temperature Parameter

(1) Red LED Temperature Compensation

The luminous intensity of typical red LEDs shows a significant temperature-dependency that is almost linear. The ID804 provides a feature to compensate this effect. For that purpose, the on-chip temperature sensor measures the chip temperature. Measured temperature value is used as a point to a lookup table. The lookup table is configured by MCU. In below figure, TC_BASE and TC_OFFSET values are stored in OTP. TEMP_LUT_TCx register value is written by MCU through SET_TEMP_LUT_TCx. The lookup table coefficient(TC) is used to RED color PWM scaling like below Fig 29 equation. For example, if the measured chip temperature value is in segment #5 in below figure, the TC value in Fig 29 is TEMP_LUT_TC5. This temperature compensation function can be enabled or disabled by SETUP1.TC_E register setting.

Color	Scaling equation
Disabled(SETUP1.TC_E=0)	$PWM_{RED} = ((PWM_{MAX}[11:0] \times R\ data[7:0] + 128) >> 8)$
Enabled(SETUP1.TC_E=1)	$PWM_{RED} = (((PWM_{MAX}[11:0] \times R\ data[7:0] + 128) >> 8) \times TC[9:0] + 256) >> 9$ $\quad \quad \quad \doteq (PWM_{RED}(When\ SETUP1.TC_E=0) \times TC[9:0] / 512) + 0.5$

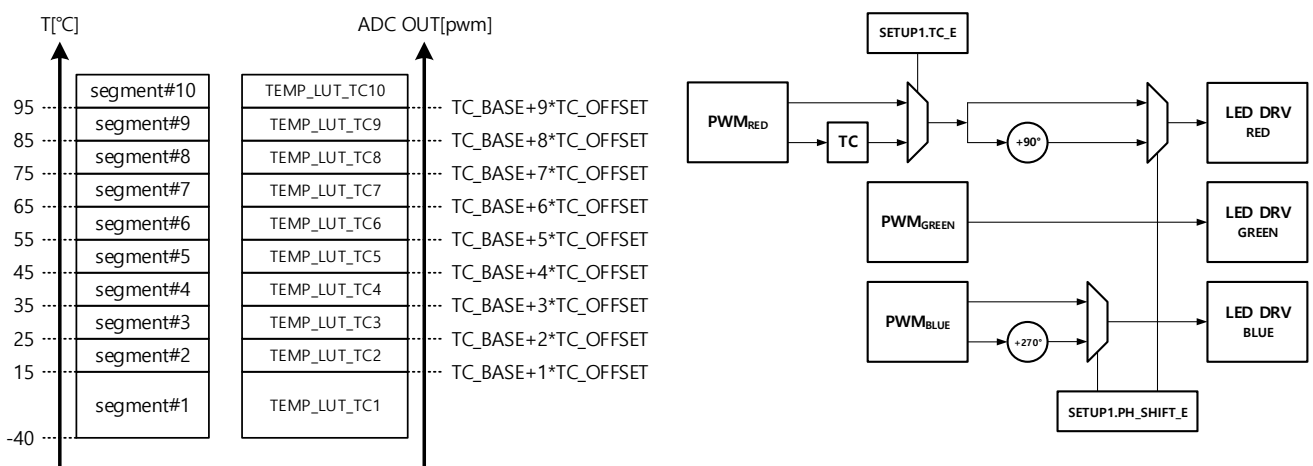


Fig 28. Scaling equation with RED Temperature Compensation Table

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24-5. Diagnostics

The ID804 provides built-in diagnostic functions for automatic detection of fault conditions.

If a fault is detected, the respective fault flag bit is set. The fault flag bit can only be cleared when the STATUS2 register has been read by the MCU (either directly or via status request bit), all flag bits have been cleared using CLRERROR, or the unit has received a RESET command. See Chapter 25-6 for more details.

If selected, the ID804 will automatically turn off the LED drivers when a failure is detected and change to SLEEP state to prevent damage to the part. This helps to ensure that communication is always possible through the entire daisy-chain.

- **Overttemperature(OT)** : This error occurs when the IC junction temperature rises above the overtemperature threshold for more than 5 consecutive EVENT_CYC cycles time. It can only be cleared once the temperature has fallen below the overtemperature release threshold. See Chapter 25-6 for details on controlling the threshold values.
- **Undervoltage(UV)** : This error occurs when the supply voltage(VDD) of the ID804 drops below the undervoltage threshold for a period longer than the deglitch time. Refer to the Electrical Characteristic for more details.

LED open / short : This fault occurs when the LED driver pin voltage remains below the LED-open threshold voltage or above the LED-short threshold voltage for 16 consecutive FPWM_DIV cycles time . These fault conditions are tested individually for each channel, but only after the GOACTIVE command has been received and the PWM value of the channel is set to a value higher than the minimum value. Refer to the Electrical Characteristic for more details.

- **Communication Error** : See Chapter 25-3.7.(6).
- **Communication Timeout** : See Chapter 25-3.7.(7)

24-5. Diagnostics

1. Fault Reaction

Fault	Detection	Fault bit	Configuration	Reaction
VDD UV	$VDD < V_{uvth_lo}$	UV_FLT	UV_FLT_E=0	Only Fault bit set
			UV_FLT_E=1	Fault bit set & LED drivers are disabled & Go to SLEEP state
OT	$T_J > \text{Temperature threshold}$	OT_FLT	OT_FLT_E=0	Only Fault bit set
			OT_FLT_E=1	Fault bit set & LED drivers are disabled & Go to SLEEP state
LED OPEN	$V_{LED} < V_{TH_OPEN}$	OPEN_R/G/B	OT_FLT_E=0	Only Fault bit set
			OT_FLT_E=1	Fault bit set & LED drivers are disabled & Go to SLEEP state
LED SHORT	$V_{LED} < V_{TH_SHORT}$	SHORT_R/G/B	OT_FLT_E=0	Only Fault bit set
			OT_FLT_E=1	Fault bit set & LED drivers are disabled & Go to SLEEP state
COMMUNICATION ERROR	See Fig 22	COM_FLT	COM_FLT_E=0	Only Fault bit set
			COM_FLT_E=1	Fault bit set & LED drivers are disabled & Go to SLEEP state
COMMUNICATION Timeout	See Fig 23	T_OUT		Only Fault bit set

Fig 29. Fault Reaction Table

24-5. Diagnostics

2. External Voltage Monitor

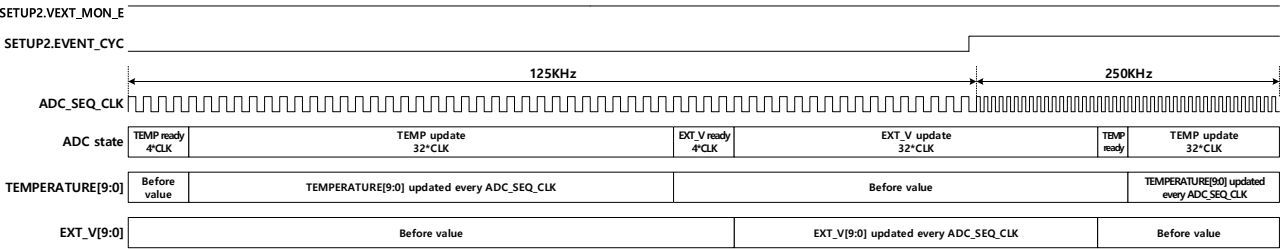
ID804 has an external voltage monitoring function using EXT_V pin. This function can be enabled by SETUP2.VEXT_MON_E=1 setting. The value of EXT_V pin voltage can be measured by 10bits ADC and it can be read from EXT_V register through READ_EXT_V command. EXT_V register value is 10'h000 when external voltage monitoring function disabled(SETUP2.VEXT_MON_E = 0).

If SETUP2.VEXT_MON_E=1, The ADC periodically measures the die temperature sensor and the EXT_V pin voltage alternately. The ADC state has a ready state for 4 ADC_SEQ_CLK periods and an update state for 32 ADC_SEQ_CLK periods. So, the temperature sensor and EXT_V pin monitor measurement time is 36 ADC_SEQ_CLK periods respectively.

The EXT_V register is updated with measured EXT_V pin voltage value every ADC_SEQ_CLK period during EXT_V update state. If SETUP2.EVENT_CYC=1, the measurement time is twice faster because the ADC_SEQ_CLK changes from 125KHz to 250KHz.

MCU can be used for touch interface application by the EXT_V register due to EXT_V pin voltage monitoring.

Parameter	Description	Equation	Unit
$V_{EXT_V,mon}$	External voltage monitor	$V_{EXT_V,mon} = 2.5 + \left(\frac{EXT_V[9:0] - d505}{d184} \right)$	V
EXT_V[9:0]	EXT_V pin voltage value converted by ADC	-	digit



24-6. ID804 Commands

The ID804 supports the following commands.

Command	Normal Command			SR (Status Request) Command			Data bits
	Encoding	Broadcast possible	Multicast possible	Encoding with SR	Broadcast possible	Multicast possible	
RESET	0xB1	O	O		-	-	0
INITBIDIR	0xB2	X	X		-	-	0
CLRERROR	0xB3	O	O	0xF3	O	X	0
GOSLEEP	0xB4	O	O	0xF4	O	X	0
GOACTIVE	0xB8	O	O	0xF8	O	X	0
GODEEPSLEEP	0xBC	O	O	0xFC	O	X	0
SET_SETUP1	0x86	O	O	0xC6	O	X	12
SET_SETUP2	0x87	O	O	0xC7	O	X	12
SET_MCAST	0x88	X	X	0xC8	X	X	24
SET_RGB	0xA0	O	O	0xE0	O	X	24
SET_TEMPTH	0x8A	O	O	0xCA	O	X	12
SET_TEMPHYS	0x8B	O	O	0xCB	O	X	12
SET_CURR_MAX_LVL	0x8F	O	O	0xCF	O	X	12
SET_TEMP_LUT_TC1	0x90	O	O	0xD0	O	X	12
SET_TEMP_LUT_TC2	0x91	O	O	0xD1	O	X	12
SET_TEMP_LUT_TC3	0x92	O	O	0xD2	O	X	12
SET_TEMP_LUT_TC4	0x93	O	O	0xD3	O	X	12
SET_TEMP_LUT_TC5	0x94	O	O	0xD4	O	X	12
SET_TEMP_LUT_TC6	0x95	O	O	0xD5	O	X	12
SET_TEMP_LUT_TC7	0x96	O	O	0xD6	O	X	12
SET_TEMP_LUT_TC8	0x97	O	O	0xD7	O	X	12
SET_TEMP_LUT_TC9	0x98	O	O	0xD8	O	X	12
SET_TEMP_LUT_TC10	0x99	O	O	0xD9	O	X	12
READ_STATUS1	0x41	O	X		-	-	12
READ_STATUS2	0x42	O	X		-	-	12
READ_TEMP	0x43	O	X		-	-	12
READ_TEMPST	0x44	O	X		-	-	24
READ_VEXT_TM	0x45	O	X		-	-	12
READ_SETUP1	0x46	O	X		-	-	12
READ_SETUP2	0x47	O	X		-	-	12
READ_MCAST	0x48	O	X		-	-	24
READ_RGB	0x60	O	X		-	-	24
READ_TEMPTH	0x4A	O	X		-	-	12
READ_TEMPHYS	0x4B	O	X		-	-	12
READ_PWM_RED_VAL	0x4C	O	X		-	-	12
READ_PWM_GREEN_VAL	0x4D	O	X		-	-	12

Fig 30. ID804 Commands

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24-6. ID804 Commands

The ID804 supports the following commands.

Command	Normal Command			SR (Status Request) Command			Data bits
	Encoding	Broadcast possible	Multicast possible	Encoding with SR	Broadcast possible	Multicast possible	
READ_PWM_BLUE_VAL	0x4E	O	X		-	-	12
READ_CURR_MAX_LVL	0x4F	O	X		-	-	12
READ_TEMP_LUT_TC1	0x50	O	X		-	-	12
READ_TEMP_LUT_TC2	0x51	O	X		-	-	12
READ_TEMP_LUT_TC3	0x52	O	X		-	-	12
READ_TEMP_LUT_TC4	0x53	O	X		-	-	12
READ_TEMP_LUT_TC5	0x54	O	X		-	-	12
READ_TEMP_LUT_TC6	0x55	O	X		-	-	12
READ_TEMP_LUT_TC7	0x56	O	X		-	-	12
READ_TEMP_LUT_TC8	0x57	O	X		-	-	12
READ_TEMP_LUT_TC9	0x58	O	X		-	-	12
READ_TEMP_LUT_TC10	0x59	O	X		-	-	12

Fig 31. ID804 Commands

If the ID804 receives a command that is not listed above, a communication error is raised.

RESET:

Performs a complete reset of one or all devices. The effect is identical to a power cycle (see Chapter 25-1). All register values are set to their default values, all error flags are cleared, the communication mode detection is restarted, LED drivers are turned off, and the address is set to 0xFFFF. The device enters the UNINIT state. After issuing the RESET command, wait for at least 1ms before sending out the next command. This command has a special characteristic for more easily re-initialize the chain, in case of some ICs have a reset by fault in the chain. If SETUP1.CRC_E = 0 configuration, whole command that received is forwarded to next IC even though received message including CRC data.

INITBIDIR:

Initiates the automatic addressing of the chain. The command shall be addressed to the first unit in the chain always with address 0x001. The last unit in the chain (indicated by the EOL mode, see Chapter 25-3.1.(2)) returns its address to the MCU. Subsequent commands should only be sent after the response has been received by the MCU. This command has a special characteristic for more easily re-initialize the chain, in case of some ICs have a reset by fault in the chain. If SETUP1.CRC_E = 1 configuration, IC make CRC data and forwarding to next IC with it, when received message has not CRC data. And command can execute even though received message has not CRC data.

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24-6. ID804 Commands

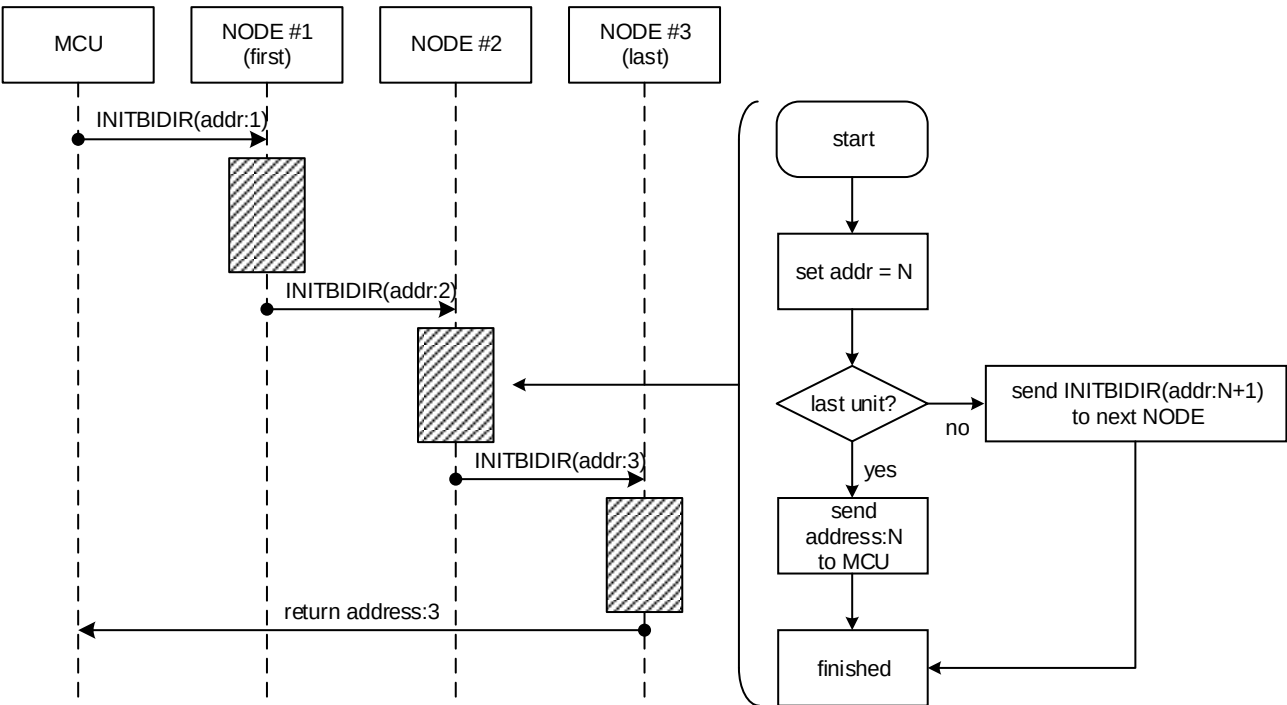


Fig 32. Automatic addressing feature

If the maximum number of node addresses has been reached (0xFEE), the unit stops the initialization procedure and returns its address (0xFEE) to the MCU.

If a node is already initialized, the command is ignored but the telegram is forwarded with incremented addresses and the last unit returns its address. This can be used to selectively initialize individual units in the chain after a RESET or POR. In this case, do not initialize the chain with an address different from 0x001 as this could cause undefined behavior.

Note that, when the last unit is not in EOL mode (e.g., CAN-FD), it will not recognize itself as last unit. Then, the application needs to ensure that the chain length is correctly received by the MCU.

Uninitialized nodes interpret this command.

Note : This command is not allowed as broadcast. If the broadcast address or multicast address is used nevertheless, all units in the chain will be initialized to address 0x000 and no response telegram will be sent. Proper operation can be restored with a broadcast RESET followed by INITBIDIR with address 0x001.

Note : If the command is sent using the broadcast address 0x000 into an already addressed chain, all units will set the COM_FLT and the last unit will return the TEMP & STATUS2 register with address 0x000.

24-6. ID804 Commands

CLRERROR:

Clears all error flags in one or all devices. If an error condition still persists, the error flag is automatically set again. If the CLRERROR with SR command received, the device returns the TEMP & STATUS2 register after the error flags have been cleared.

GOSLEEP : Sends one or all devices into SLEEP state. See Chapter 26-1.

GOACTIVE : Sends one or all devices into ACTIVE state. See Chapter 26-1.

GODEEPSLEEP : Sends one or all devices into DEEPSLEEP state. Chapter 26-1.

SET_SETUP1 / READ_SETUP1 : Write / Return the SETUP1 register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	TC_E	OS_FLT_E	OT_FLT_E	UV_FLT_E	COM_FLT_E	CRC_E	CLK_INV	LG_E	F_PWM_DIV[3:0]			
Default	0	0	0	0	0	0	0	0	0x0			
Access	RW	RW	RW	RW	RW	RW	RW	RW	RW			
TC_E			0: RED LED temperature compensation disable. 1: RED LED temperature compensation enable.									
OS_FLT_E			0: LED open/short fault flag, OS_FLT, is raised. 1: OS_FLT bit is raised, and LED drivers are disabled.									
OT_FLT_E			0: Overtemperature fault flag, OT_FLT, is raised. 1: OT_FLT bit is raised, and LED drivers are disabled.									
UV_FLT_E			0: Undervoltage fault flag, UV_FLT, is raised. 1: UV_FLT bit is raised, and LED drivers are disabled.									
COM_FLT_E			0: Communication fault flag, C_FLT, is raised. 1: C_FLT bit is raised, and LED drivers are disabled.									
CRC_E			0: CRC disabled. 1: CRC enabled.									
CLK_INV			CLK polarity for MCU mode 1 : Inverted									
LG_E			0: Low Grayscale PWM disable. 1: Low Grayscale PWM enable. If RGB data ≤ 0x8, the PWM frequency is FPWM /4.									
F_PWM_DIV[3:0]			PWM frequency divider. The PWM Frequency is activated in only active mode. FPWM = mclk / (F_PWM_DIV+1)									

24-6. ID804 Commands

SET_SETUP2 / READ_SETUP2 : Write / Return the SETUP2 register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name							VEXT_MON_E	EVENT_CYC	UV_LVL[1:0]		SH_LVL[1:0]	
Default							0	0	0x11		0x10	
Access							RW	RW	RW		RW	
VEXT_MON_E	0: EXT_TM pin voltage monitoring disable. 1: EXT_TM pin voltage monitoring enable.											
EVENT_CYC	This bit determines a period to detect fault events or to update the ADC output value. If VEXT_MON_E '1', then the ADC update time will be doubled. 0: 15.625khz (typical 64us) 1: 31.25khz (typical 32us)											
UV_LVL[1:0]	VDD UVLO detection level. 2'b00 : VUV_TH_Rise = 3.65V, VUV_TH_Fall = 3.50V 2'b01 : VUV_TH_Rise = 3.80V, VUV_TH_Fall = 3.65V 2'b10 : VUV_TH_Rise = 4.00V, VUV_TH_Fall = 3.80V 2'b11 : VUV_TH_Rise = 4.20V, VUV_TH_Fall = 4.00V (default)											
SH_LVL[1:0]	LED Short detection level. 2'b00 : VS_TH = 3.75V. 2'b01 : VS_TH = 4.00V. 2'b10 : VS_TH = 4.25V. (default) 2'b11 : VS_TH = 4.50V.											

24-6. ID804 Commands

SET_MCAST / READ_MCAST : Write / Return the MCAST register

Bit	[23]	[22]	[21]	[20]	[19]	[18]	[17]	[16]	[15]	[14]	[13]	[12]
Name									MCAST_ADDR [15:12]			
Default									0x0			
Access									RW			
Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	MCAST_ADDR [11:0]											
Default	0x00											
Access	RW											

MCAST_ADDR [15:0]

Multicast group address.

In case that the device address indicates the multicast group address.

The device can be assigned to multiple of the 16 multicast groups.

All devices received the command act like a broadcast and all device belonging to a multicast address should treat the command transaction.

This register can be accessed by SET_MCAST and READ_MCAST command.

[0] = 1 : Multicast group address 0xFE0

[1] = 1 : Multicast group address 0xFF0

□□□

[15] = 1 : Multicast group address 0xFFE

12 bits device addr

0x000 → broad-cast addr

0x001~0xFEE → individual node addr

0xFE0~0xFFE → multi-cast addr

0xFFF → default addr

24-6. ID804 Commands

SET_RGB / READ_RGB : Write / Return the RGB register

Bit	[23]	[22]	[21]	[20]	[19]	[18]	[17]	[16]	[15]	[14]	[13]	[12]
Name	R_DATA [7:0]								G_DATA [7:4]			
Default	0x00								0x00			
Access	RW								RW			
Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	G_DATA [3:0]				B_DATA [7:0]							
Default	0x00				0x00							
Access	RW				RW							
R_DATA [7:0]			Red color data R_PWM data = (R_DATA x PWMMAX [*]) + 128) >> 8 (when SETUP1.TC_E=0) R_PWM data = [{(R_DATA x PWMMAX [*]) + 128) >> 8} x TC +256] >> 9 (when SETUP1.TC_E=1)									
G_DATA [7:0]			Green color data This data is converted into 12-bit PWM data toward a sink driver. G_PWM data = (G_DATA x PWMMAX [*]) + 128) >> 8									
B_DATA [7:0]			This data is converted into 12-bit PWM data toward a sink driver. B_PWM data = (B_DATA x PWMMAX [*]) + 128) >> 8									

SET_TEMPHTH / READ_TEMPHTH : Write / Return the TEMPHTH register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	TEMP_THRESHOLD[9:0]											
Default	0x000 (Initial value setting required)											
Access	RW											
TEMP_THRESHOLD[9:0]			Overtemperature fault detection threshold. The OT_FLT is raised automatically the temperature increases above this value. If TEMP_THRESHOLD is set to 0x000, the OT diagnosis is disabled. To use the OT monitoring function, TEMP_THRESHOLD[9:0] value other than 0 must be set. TEMP_THRESHOLD[9:0] = (T _{J_OT} - 25°C)x3.6 + 512									

24-6. ID804 Commands

SET_TEMP_PHYS / READ_TEMP_PHYS : Write / Return the TEMP_PHYS register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name			TEMP_HYSTERISIS[9:0]									
Default			0x000 (Initial value setting required)									
Access			RW									
TEMP_HYSTERISIS[9:0]			Overtemperature fault detection threshold. The OT_FLT can be cleared only when the temperature is below this value. The TEMP_HYSTERISIS setting value must be set smaller than TEMP_THRESHOLD. Otherwise, OT diagnosis will not work.									

SET_TEMP_LUT_TC1 / READ_TEMP_LUT_TC1 : Write / Return the TEMP_LUT_TC1 register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	TEMP_LUT_TC1[9:0]											
Default	0x200											
Access	RW											
TEMP_LUT_TC1[11:0]			10bit TEMP LOOK-UP-TABLE for Temperature Compensation rate Segment #1 Temperature compensation range is from -40 to 15°C. The default Compensation rate for segment #1 is 0%. “(0x200/0x200)-1” The compensation rate is applied only to the red channel based on the die temperature value read by the ADC.									

24-6. ID804 Commands

SET_TEMP_LUT_TC2 / READ_TEMP_LUT_TC2 : Write / Return the TEMP_LUT_TC2 register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name			TEMP_LUT_TC2[11:0]									
Default			0x200									
Access			RW									
TEMP_LUT_TC2[11:0]			10bit TEMP LOOK-UP-TABLE for Temperature Compensation rate Segment #2 Temperature compensation range is from 15 to (15+10)°C. The default Compensation rate for segment #2 is 0%. “(0x200/0x200)-1” The compensation rate is applied only to the red channel based on the die temperature value read by the ADC.									

SET_TEMP_LUT_TC3 / READ_TEMP_LUT_TC3 : Write / Return the TEMP_LUT_TC3 register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name			TEMP_LUT_TC3[11:0]									
Default			0x21A									
Access			RW									
TEMP_LUT_TC3[11:0]			10bit TEMP LOOK-UP-TABLE for Temperature Compensation rate Segment #3 Temperature compensation range is from 25 to (25+10)°C. The default Compensation rate for segment #3 is +5%. “1-(0x21A/0x200)-1” The compensation rate is applied only to the red channel based on the die temperature value read by the ADC.									

SET_TEMP_LUT_TC4 / READ_TEMP_LUT_TC4 : Write / Return the TEMP_LUT_TC4 register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name			TEMP_LUT_TC4[11:0]									
Default			0x231									
Access			RW									
TEMP_LUT_TC4[11:0]			10bit TEMP LOOK-UP-TABLE for Temperature Compensation rate Segment #4 Temperature compensation range is from 35 to (35+10)°C. The default Compensation rate for segment #4 is +9.5%. “(0x231/0x200)-1” The compensation rate is applied only to the red channel based on the die temperature value read by the ADC.									

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24-6. ID804 Commands

SET_TEMP_LUT_TC5 / READ_TEMP_LUT_TC5 : Write / Return the TEMP_LUT_TC5 register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name			TEMP_LUT_TC5[11:0]									
Default			0x248									
Access			RW									
TEMP_LUT_TC5[11:0]			10bit TEMP LOOK-UP-TABLE for Temperature Compensation rate Segment #5 Temperature compensation range is from 45 to (45+10)°C. The default Compensation rate for segment #5 is +14%. “(0x248/0x200)-1” The compensation rate is applied only to the red channel based on the die temperature value read by the ADC.									

SET_TEMP_LUT_TC6 / READ_TEMP_LUT_TC6 : Write / Return the TEMP_LUT_TC6 register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name			TEMP_LUT_TC6[11:0]									
Default			0x25F									
Access			RW									
TEMP_LUT_TC6[11:0]			10bit TEMP LOOK-UP-TABLE for Temperature Compensation rate Segment #6 Temperature compensation range is from 55 to (55+10)°C. The default Compensation rate for segment #6 is +18.5%. “(0x25F/0x200)-1” The compensation rate is applied only to the red channel based on the die temperature value read by the ADC.									

SET_TEMP_LUT_TC7 / READ_TEMP_LUT_TC7 : Write / Return the TEMP_LUT_TC7 register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name			TEMP_LUT_TC7[11:0]									
Default			0x276									
Access			RW									
TEMP_LUT_TC7[11:0]			10bit TEMP LOOK-UP-TABLE for Temperature Compensation rate Segment #7 Temperature compensation range is from 65 to (65+10)°C. The default Compensation rate for segment #7 is +23%. “(0x276/0x200)-1” The compensation rate is applied only to the red channel based on the die temperature value read by the ADC.									

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24-6. ID804 Commands

SET_TEMP_LUT_TC8 / READ_TEMP_LUT_TC8 : Write / Return the TEMP_LUT_TC8 register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name			TEMP_LUT_TC8[11:0]									
Default			0x28D									
Access			RW									
TEMP_LUT_TC8[11:0]			10bit TEMP LOOK-UP-TABLE for Temperature Compensation rate Segment #8 Temperature compensation range is from 75 to (75+10)°C. The default Compensation rate for segment #8 is +27.5%. "(0x28D/0x200)-1" The compensation rate is applied only to the red channel based on the die temperature value read by the ADC.									

SET_TEMP_LUT_TC9 / READ_TEMP_LUT_TC9 : Write / Return the TEMP_LUT_TC9 register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name			TEMP_LUT_TC9[11:0]									
Default			0x2A4									
Access			RW									
TEMP_LUT_TC9[11:0]			10bit TEMP LOOK-UP-TABLE for Temperature Compensation rate Segment #9 Temperature compensation range is from 85 to (85+10)°C. The default Compensation rate for segment #9 is +32%. "(0x2A4/0x200)-1" The compensation rate is applied only to the red channel based on the die temperature value read by the ADC.									

SET_TEMP_LUT_TC10 / READ_TEMP_LUT_TC10 : Write / Return the TEMP_LUT_TC10 register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name			TEMP_LUT_TC10[11:0]									
Default			0x2BB									
Access			RW									
TEMP_LUT_TC10[11:0]			10bit TEMP LOOK-UP-TABLE for Temperature Compensation rate Segment #10 Temperature compensation range is from 95°C. The default Compensation rate for segment #10 is +36.5%. "(0x2BB/0x200)-1" The compensation rate is applied only to the red channel based on the die temperature value read by the ADC.									

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24-6. ID804 Commands

SET_CURR_MAX_LVL / READ_CURR_MAX_LVL: Write / Return the CURR_MAX_LVL register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	R_CURR_MAX_LVL[3:0]				G_CURR_MAX_LVL[3:0]				B_CURR_MAX_LVL[3:0]			
Default	0x8				0x8				0x8			
Access	RW				RW				RW			

				This determines the max current value of the output each (R/G/B) sink driver.								
				4'b0000: 1.45mA								
				4'b0001: 2.90mA								
				4'b0010: 4.35mA								
				4'b0011: 5.80 mA								
				4'b0100: 7.24mA								
				4'b0101: 8.69mA								
				4'b0110: 10.14mA								
R/G/B_CURR_MAX_LVL[3:0]				4'b0111: 11.58mA								
				4'b1000: 13.03mA								
				4'b1001: 14.47mA								
				4'b1010: 15.92mA								
				4'b1011: 17.36mA								
				4'b1100: 18.81mA								
				4'b1101: 20.25mA								
				4'b1110: 21.70mA								
				4'b1111: 23.14mA								

SET_TIMEOUT / READ_TIMEOUT: Write / Return the TIMEOUT register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	TIMEOUT [11:0]											
Default	0x800											
Access	RW											

TIMEOUT [11:0]				Communication timeout length set. TIMEOUT 1LSB : mclk(62.5ns) * 2^10 = 64us Default TIMEOUT = (0x800)*(64us) = 131ms								
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24-6. ID804 Commands

READ_STATUS1 : Return the STATUS1 register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name						IC_STATE[2:0]			SIO2[1:0]		SIO1[1:0]	
Default						0x0			0x0		0x0	
Access						RO			RO		RO	
IC_STATE[2:0]			Represent current IC state. 3'b000 : POR. (default) 3'b001 : Uninitialized. 3'b010 : Sleep. 3'b011 : Active. 3'b100 : Deepsleep.									
SIO2[1:0]			Communication mode of SIO2. 2'b00 : LVDS. (default) 2'b01 : EOL. 2'b10 : MCU 2'b11 : CAN Value is set automatically at startup and after reset.									
SIO1[1:0]			Communication mode of SIO1. 2'b00 : LVDS. (default) 2'b01 : EOL. 2'b10 : MCU 2'b11 : CAN Value is set automatically at startup and after reset.									

24-6. ID804 Commands

READ_STATUS2 : Return the STATUS2 register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	OTPCRC	T_OUT	OPEN_R	OPEN_G	OPEN_B	SHORT_R	SHORT_G	SHORT_B	OT_FLT	UV_FLT	CRC_FLT	COM_FLT
Default	0	0	0	0	0	0	0	0	0	0	0	0
Access	RC	RC	RC	RC	RC	RC	RC	RC	RC	RC	RC	RC
OTPCRC			OTP error flag set during startup when a mismatch between the OTP values and the internal CRC is detected. Cleared after readout									
T_OUT			Frame Time Out event flag is raised if the response is not arrived within a fixed time, and auto-cleared after readout.									
OPEN_R			Red LED Open event flag is raised if $V_{HD} \leq V_{OPEN}$ for more than T_{diag_filter} time and the PWM data is not zero (not R/G/B DATA), and auto-cleared after readout. If the event flag is raised under OS_FLT_E '1', then the LED sink driver is turned off.									
OPEN_G			Green LED Open event flag is raised $V_{LED, GREEN} < V_{TH_OPEN}$ for more than T_{diag_filter} time and the PWM output signal is active. If the event flag is raised under OS_FLT_E '1', then the device state will transit to a SLEEP state and the LED sink driver will be turned off. This bit is auto-cleared after readout, but if the detection condition is still valid after readout, this bit will be set again immediately unless the condition is removed.									
OPEN_B			Blue LED Open event flag is raised if $V_{LED, BLUE} < V_{TH_OPEN}$ for more than T_{diag_filter} time and the PWM output signal is active. If the event flag is raised under OS_FLT_E '1', then the device state will transit to a SLEEP state and the LED sink driver will be turned off. This bit is auto-cleared after readout, but if the detection condition is still valid after readout, this bit will be set again immediately unless the condition is removed.									

24-6. ID804 Commands

READ_STATUS2 : Return the STATUS2 register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	OTPCRC	T_OUT	OPEN_R	OPEN_G	OPEN_B	SHORT_R	SHORT_G	SHORT_B	OT_FLT	UV_FLT	CRC_FLT	COM_FLT
Default	0	0	0	0	0	0	0	0	0	0	0	0
Access	RC	RC	RC	RC	RC	RC	RC	RC	RC	RC	RC	RC
SHORT_R	Red LED Short event flag is raised if $V_{LED,RED} > V_{SHORT_TH}$ for more than T_{diag_filter} time and the PWM output signal is active. If the event flag is raised under OS_FLT_E '1', then the device state will transit to a SLEEP state and the LED sink driver will be turned off. This bit is auto-cleared after readout, but if the detection condition is still valid after readout, this bit will be set again immediately unless the condition is removed.											
SHORT_G	Green LED Short event flag is raised if $V_{LED,GREEN} > V_{SHORT_TH}$ for more than T_{diag_filter} time and the PWM output signal is active. If the event flag is raised under OS_FLT_E '1', then the device state will transit to a SLEEP state and the LED sink driver will be turned off. This bit is auto-cleared after readout, but if the detection condition is still valid after readout, this bit will be set again immediately unless the condition is removed.											
SHORT_B	Blue LED Short event flag is raised if $V_{LED,BLUE} > V_{SHORT_TH}$ for more than T_{diag_filter} time and the PWM output signal is active. If the event flag is raised under OS_FLT_E '1', then the device state will transit to a SLEEP state and the LED sink driver will be turned off. This bit is auto-cleared after readout, but if the detection condition is still valid after readout, this bit will be set again immediately unless the condition is removed.											
OT_FLT	Over temperature fault flag is raised if $TEMPERATURE[9:0] > TEMP_THRESHOLD[9:0]$ for more than 5 consecutive EVENT_CYC cycles time. If the event flag is raised under OT_FLT_E '1', then the device state will transit to a SLEEP state and the LED sink driver will be turned off. This bit is auto-cleared after readout in $TEMPERATURE[9:0]$											
UV_FLT	Undervoltage fault flag is raised if $VDD < V_{uvth_lo}$ for more than 5 consecutive EVENT_CYC cycles time. If the event flag is raised under UV_FLT_E '1', then the device state will transit to a SLEEP state and the LED sink driver will be turned off. This bit is auto-cleared after readout, but if the detection condition is still valid after readout, this bit will be set again immediately unless the condition is removed.											

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24-6. ID804 Commands

READ_STATUS2 : Return the STATUS2 register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	OTPCRC	T_OUT	OPEN_R	OPEN_G	OPEN_B	SHORT_R	SHORT_G	SHORT_B	OT_FLT	UV_FLT	CRC_FLT	COM_FLT
Default	0	0	0	0	0	0	0	0	0	0	0	0
Access	RC	RC	RC	RC	RC	RC	RC	RC	RC	RC	RC	RC
CRC_FLT			CRC fault flag is raised if CRC is wrong under CRC_E '1', and auto-cleared after readout.									
COM_FLT			Communication fault flag is raised if any communication error is occurred, and auto-cleared after readout. If the fault flag is raised under COM_FLT_E '1', then all LED sink drivers are turned off.									

READ_TEMP : Return the TEMP register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	TEMPERATURE[9:0]											
Default	0x000											
Access	RO											
TEMPERATURE[9:0]			The die temperature sensor value is converted through 10bit ADC.									

24-6. ID804 Commands

READ_EXT_V : Return the EXT_V register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	VEXT_TM[9:0]											
Default	0x000											
Access	RO											
VEXT_TM[9:0]			The external-voltage value is converted through 10bit ADC.									

READ_PWM_RED_VAL : Return the PWM_RED_VAL register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	PWM_RED_VAL[11:0]											
Default	0x0000											
Access	RO											
PWM_RED_VAL[11:0]			12bit RED PWM calculated data value									

READ_PWM_GREEN_VAL : Return the PWM_GREEN_VAL register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	PWM_GREEN_VAL[11:0]											
Default	0x0000											
Access	RO											
PWM_GREEN_VAL[11:0]			12bit GREEN PWM calculated data value									

READ_PWM_BLUE_VAL : Return the PWM_BLUE_VAL register

Bit	[11]	[10]	[9]	[8]	[7]	[6]	[5]	[4]	[3]	[2]	[1]	[0]
Name	PWM_BLUE_VAL[11:0]											
Default	0x0000											
Access	RO											
PWM_BLUE_VAL[11:0]			12bit BLUE PWM calculated data value									

25-1. Half-Duplex Communication

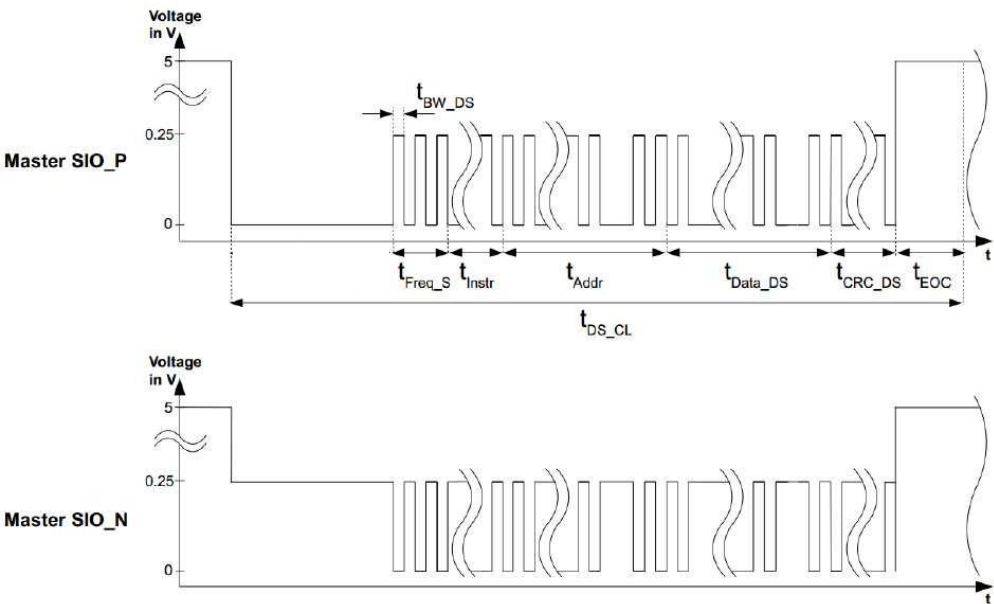


Fig 33. Differential downstream command frame

Downstream command frame parameters

Symbol	Description	Min	Typ	Max	Unit
t_{BW_DS}	Downstream bit width	384	500	714	ns
t_{DS_CL}	Downstream command length CRC enabled CRC disabled		$86 \times t_{BW_DS}$ $76 \times t_{BW_DS}$		ns ns
t_{Freq_S}	Frequency sync		$5 \times t_{BW_DS}$		ns
T_{inst}	Instruction		$5 \times t_{BW_DS}$		ns
t_{Addr}	Address		$15 \times t_{BW_DS}$		ns
t_{Data_DS}	Downstream data		$30 \times t_{BW_DS}$		ns
t_{CRC_DS}	CRC downstream		$10 \times t_{BW_DS}$		ns
t_{EOC}	End of command idle		$6 \times t_{BW_DS}$		ns

Table 19. Downstream command frame parameters

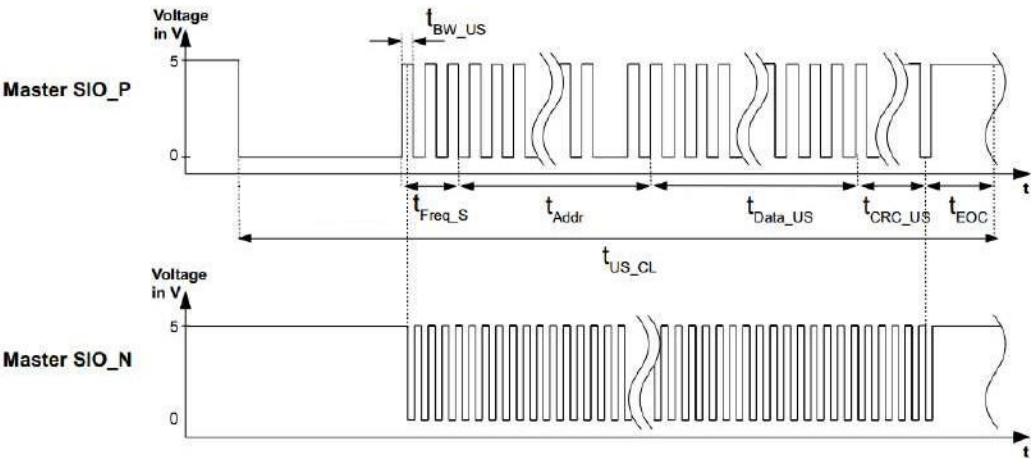


Fig 34. Single ended upstream response frame

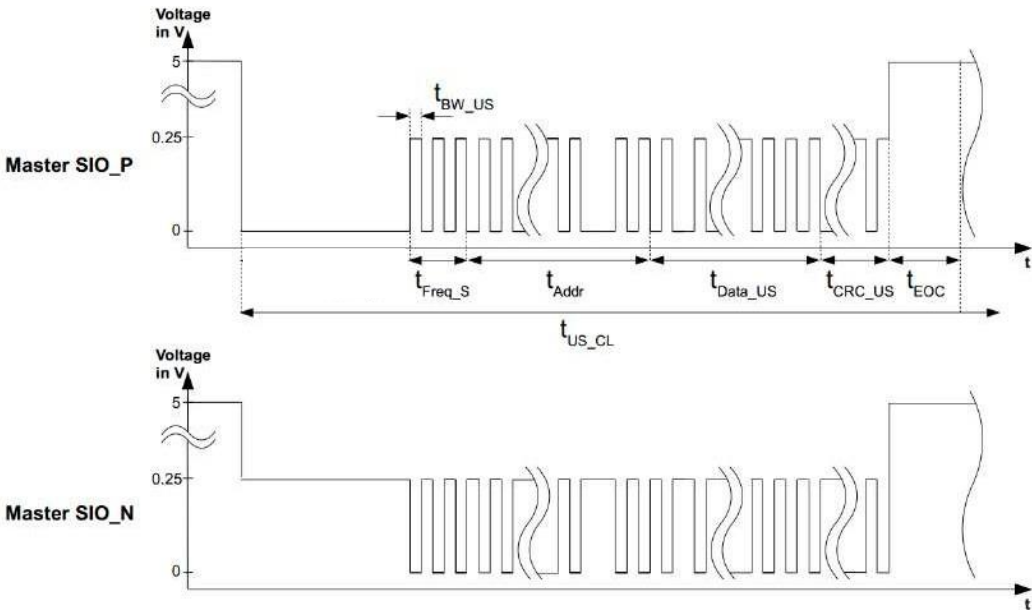


Fig 35. Differential upstream response frame

25-2. Upstream command frame parameters

Symbol	Description	Min	Typ	Max	Unit
t_{BW_US}	Upstream bit width	384	500	714	ns
t_{US_CL}	Upstream command length				
	CRC enabled		$61 \times t_{BW_DS}$		ns
	CRC disabled		$56 \times t_{BW_DS}$		ns
t_{Freq_S}	Frequency sync		$5 \times t_{BW_DS}$		ns
t_{Addr}	Address		$15 \times t_{BW_DS}$		ns
t_{Data_US}	Upstream data		$15 \times t_{BW_DS}$		ns
t_{CRC_US}	CRC downstream		$5 \times t_{BW_DS}$		ns
t_{EOC}	End of command idle		$6 \times t_{BW_DS}$		ns

Table 21. Upstream command frame parameters

25-3. Bit Retransmission

To ensure a correct bit-timing, the forwarded data is regenerated with the clock of the device.

A new frequency synchronization is only created after the freq-sync on the reception side has been received (the first four bits). Therefore, the minimum propagation delay t_{pd} introduced by the retransmission is four bit widths t_{BW} .

26-4. Propagation delay

Symbol	Description	Min	Typ	Max	Unit
t_{pd}	Propagation delay	2.0	4.0	5.2	μs

Table 20. Propagation delay

To guarantee a correct bit-timing the device uses its own clock as reference and will never transmit faster than its own bit-timing defines, but if the received freq-sync was slower, this timing is used for the retransmission.

The retransmission uses a FIFO to compensate for speed differences between reception and transmission.

Due to the variance in the oscillator clocks of different devices, after each transmission a pause of 43% of the nominal transmission time has to be introduced. If the transmission is created by a chip with $\pm 30\%$ oscillator clock variation the time has to be increased to a total of 70% of the transmission duration.

25-5. Initialization

The digLED_Init_Strip command initializes a particular ID804 chain by issuing the command on an Associated ID804 communication channel.

This command is always the first command to be transmitted after power-up or reset.

The command initializes a chain of devices by assigning the address of the device and by en- or disabling the phaseshift, the CRC and temperature compensation functions. The digLED_Init_Strip command is always executed with a CRC checksum. This is true for both, the command and the response frame.

If any command is received by a device before initialization, the command is always considered as illegal and the error status bit for an undefined command is set. This may happen in the chain's first device only, as a non-initialized device does not forward received messages.

If the first device in the chain receives a digLED_Init_Strip command, it takes the received address as its own device address and afterwards transmits another digLED_Init_Strip frame to the next device in the chain.

It increments the address before the transmission. As the adjacent devices proceed in the same manner, the devices in the chain get enumerated with ascending addresses. When the final device in the chain recognizes there is no receiving device at its downstream link, it transmits a response frame upstream. The response frame to a digLED_Init_Strip command carries the configuration word read from the OTP. It also transmits the own devices address just initialized.

All upstream devices wait for the responses to be received and forward them towards the microcontroller.

If a frame with an address equal to the adjacent device address (own address plus one) is received, the own response to the digLED_Init_Strip command is transmitted thereafter. If the first device has transmitted its response frame, the chain is ready to process regular commands (non-Init frames).

As soon as a device is initialized, it unconditionally forwards incoming correct frames (Freq-Sync and the RLC coding as well as the frame length are checked) to the adjacent node in the chain.

Frames received from upstream are forwarded downstream and vice versa. If an error is detected, the forwarding is stopped for this frame.

Symbol	Description	Equation
t _{INIT}	Initialization duration	$n \times (t_{DS_CL} + t_{US_CL} + 2 \times t_{PD})$

Table 23. Initialization duration

25-6. Write Access

Most commands of the LED Controller are write-only commands. I.e. the devices receive a command frame and execute the appropriate actions without any further communication. A write access command may be directed to a single device (unicast), to all devices (broadcast), or to a defined group of devices (multicast).

As every command frame is forwarded downstream irrespective of its destination address, all stations always receive all commands. Only its execution depends on the command's destination address. To avoid communication issues, it is recommended to wait 30% of the command length between two consecutive commands.

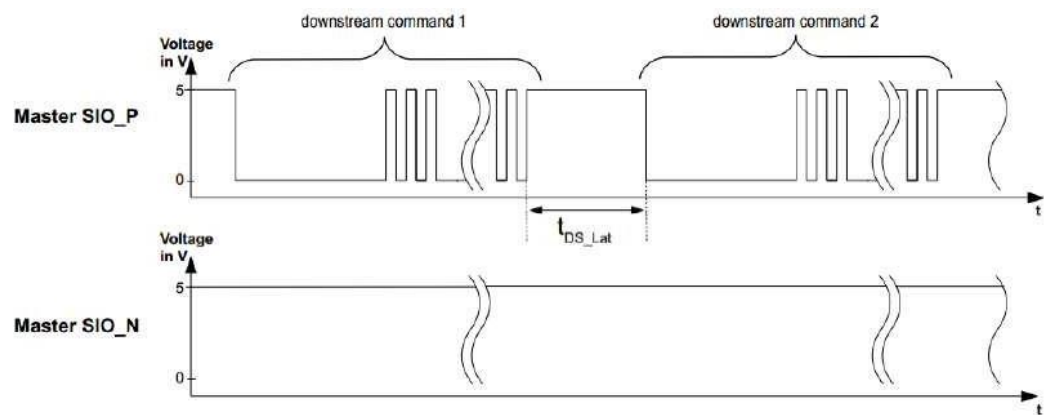


Fig 36. Multiple single ended downstream commands

Symbol	Description	Min.
t_{DS_Lat}	Latency between two downstream commands	$0.3 \times t_{DS_CL}$

Table 21. Recommended latency between downstream commands

25-7. Read Access

A read access consists of two phases, the command and the response phase. The command phase uses downstream communication and the response phase uses upstream communication. Commands for read access do not use the command address, i.e. these commands may not be directed to a device based on the device address.

There are two commands for read access, digLED_Read and digLED_Ping. The digLED_Read commands retrieve a status information from all devices and the digLED_Ping command is used to check the device chain's integrity. Only the final node in the chain responds to a PING command.

A digLED_Read command is first received by all devices via the frame in downstream direction.

The last node in the chain then immediately transmits its response frame upstream. The response frame's data field depends on the actual digLED_Read command. The response frame's address field is set according to the own device's address. All the nodes upstream forward all received response frames until a frame with the address of their adjacent node is received. Then the respective node transmits its own response frame.

This procedure lasts until the chain's first node has transmitted its response frame.

A digLED_Ping command is similar to a digLED_Read command, but only the last device in the chain responds to a digLED_Ping. Thus, the digLED_Ping command is executed much faster than a regular digLED_Read command.

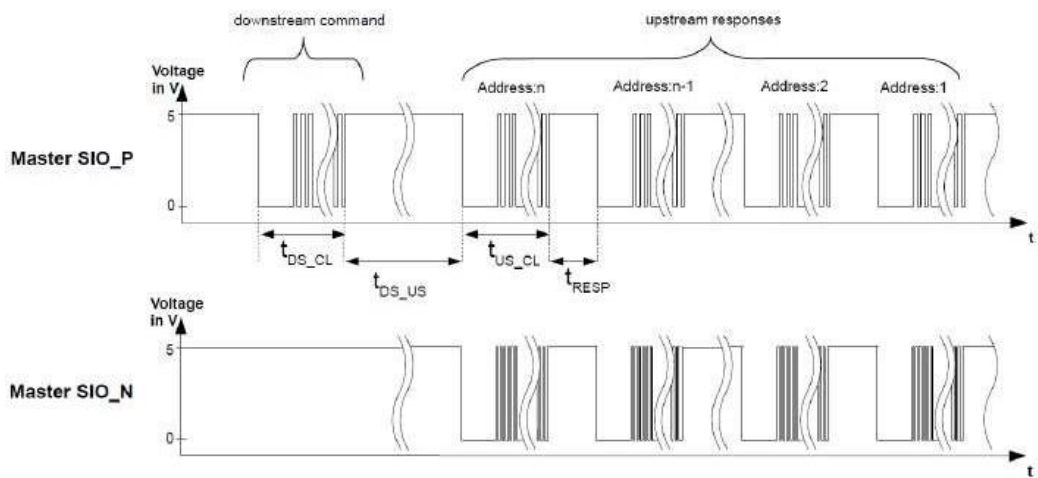


Fig 37. Single ended read command and responses

Down- and upstream delay, delay between responses

Symbol	Description	Min.
t_{DS_US}	Delay between down- and upstream	$t_{DS_CL} + t_{US_CL} + 2 \times n \times t_{PD}$
t_{RESP}	Delay between responses Oscillator variation of adjacent devices < ±30% Oscillator variation of adjacent devices > ±30%	$0.43 \times t_{US_CL}$ $0.7 \times t_{US_CL}$

Table 25. Down- and upstream delay, delay between responses

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25-8. Timeouts

The digLED_Init_Strip, all the digLED_Read, and the digLED_Ping commands initiate upstream data transmission. With the digLED_Init_Strip and the read commands all nodes are expected to send a response to the host. The digLED_Ping requires only the last node in the LED chain to respond. However, in all cases each node needs to await all responses originating from the nodes downstream. Thereafter either the node's own response is transmitted or new commands are accepted. Only the last node in the LED chain may immediately transmit its response.

In case there is an error with the chain downstream, not all expected responses may arrive. Thus, each of the commands expecting a response waits for a certain time only and then returns to its previous state without having transmitted the node's response data.

The lengths of the timeouts depend on the respective command. They are calculated to account for the worst-case oscillator frequency tolerance. I.e. the waiting node has a high-speed clock and all the nodes waited for have a low speed clock. The hardware implementation uses an internally divided clock for the timeout counter:

$$f_{[timeout]} = f_{[osc]} / 2^{14}$$

With the nominal clock frequency of 16MHz the counter's resolution results to 1.024ms.

Timeouts

Command	Max. Counter value	timeout			Unit
		Min	Nom.	Max	
digLED_Init_Strip	992	780.6	1015.3	1451.2	ms
digLED_Read_*	427	335.6	436.7	624.6	ms
digLED_Ping	62	48.0	63.0	90.7	ms

Table 22. Timeouts

25-9. PWM Units

Basic Mode of Operation

The LED controller device incorporates three independent PWM channels, one for each LED.

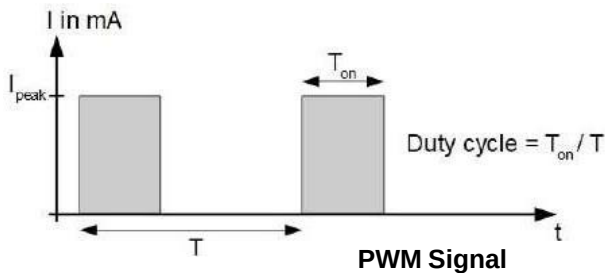


Fig 38. PWM Signal

The resolution is 12 bit. The supported duty cycles are 0/4095 to 4095/4095. The nominal PWM output frequency is $16\text{MHz} \div 215 = 488.3\text{Hz}$. The frequency is reduced to the half or the quarter of this frequency with low duty cycles. This ensures a minimum on-time of $2\mu\text{s}$ for the LEDs. The minimum output frequency is 122Hz. The output frequency is not derived from the actual PWM duty cycle but from the RGB value received from the host. As the DIM command also has impact to the LED intensity, it is accounted for as well. The actual relationship is given in the following table.

DIM parameter relationships between RGB parameters and PWM frequencies

DIM Parameter	RGB Parameter	PWM Frequency in Hz
0	8...255	488
	4...7	244
	0...3	122
1	16...255	488
	8...15	244
	0...7	122
2	32...255	488
	16...31	244
	0...15	122
3	64...255	488
	32...63	244
	0...31	122

Table 23. DIM parameter relationships between RGB parameters and PWM frequencies

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Update

When a new PWM duty cycle has to be applied, this is always done at the end of a PWM cycle. I.e. the PWM always completes an output cycle using the previously active duty cycle and starts the next output cycle using the updated duty cycle.

Phase Shift

In order to spread the current consumption of the LEDs over time, a phase shift can be set between the three PWM channels. This optional function can be enabled/disabled during device initialization.

If the phase shift is deactivated, the red channel controls all three outputs and thus provides the temperature compensation function for all three channels.

If the phase shift is enabled, it retains even if the output frequency of the channels is different. If a channel is operating at a lower frequency, it may be considered to leave out one or three full PWM cycles. When leaving power save mode, the channels are restarted appropriately to again obtain the correct phase shift.

The fixed phase shift is defined in the following table. Please note the absolute phase shift times are nominal values. I.e. they are subject to vary with the internal oscillator's frequency.

Phase Shift

DIM channel	Rel. Phase Shift
Green	0%
Red	25%
Blue	75%

Table 24. Phase Shift

Power Save Mode

When all LED channels are set to an intensity of 0, the device enters a power save mode for the current sources driving the LEDs. I.e. the digLED_Set_RGB command must be issued with an RGB value of 0x000000 to enter the power save mode.

Recovering from this mode does not require any particular measures. I.e. the host just needs to issue a digLED_Set_RGB command with the data field different from 0x000000 and the current sources are restarted again. There is a delay of approx. 1µs before the restart of the green PWM channel (no phase shift applies to the green channel).

This is due to an internal ramp-up required by the analog circuitry. The same procedure is applied after device power-up or a hardware reset, as the initial RGB value is 0x000000. I.e. the LEDs are all turned off after power-up or a hardware reset.

25-10. DIM Function for Accurate Low Light Colors

To extend the SET_RGB command’s resolution for accurate low light colors, the Global Technologies LED provides the DIM command. The command divides the PWM duty cycles computed from the RGB setting. There are four divisors available. Details are shown in Figure and table.

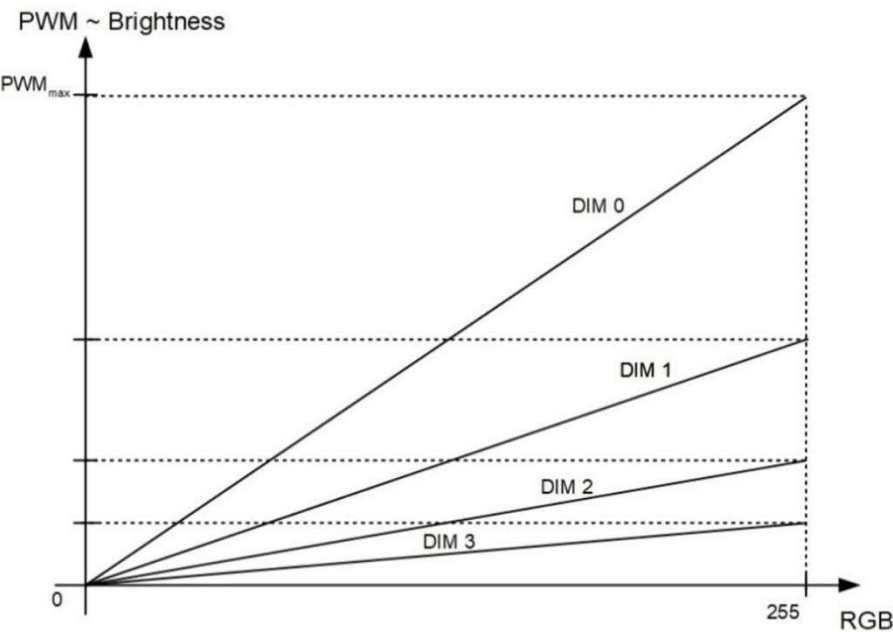


Fig 39. DIM Function for Accurate Low Light Colors

DIM Function

DIM	Relative PWM Ratio
0	1
1	1/2
2	1/4
3	1/8

Table 25. DIM Function

26. Typical Electrical Optical Characteristics Curves

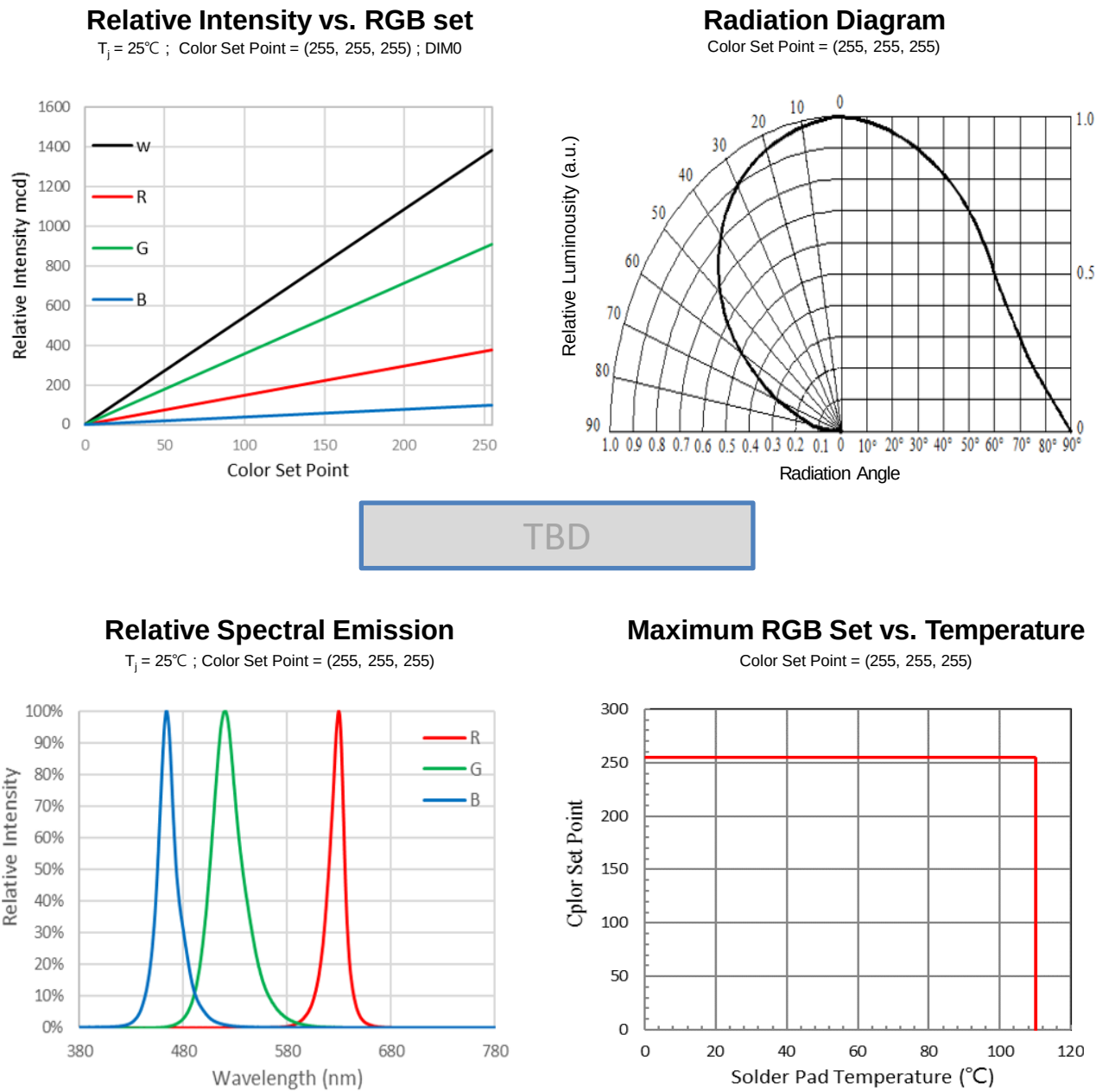
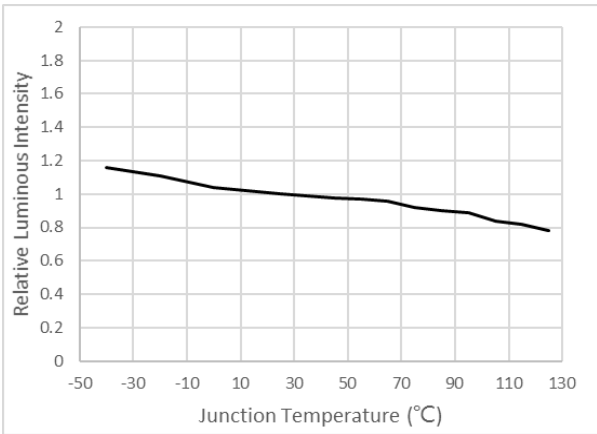


Fig 40. Typical Electrical Optical Characteristics Curves

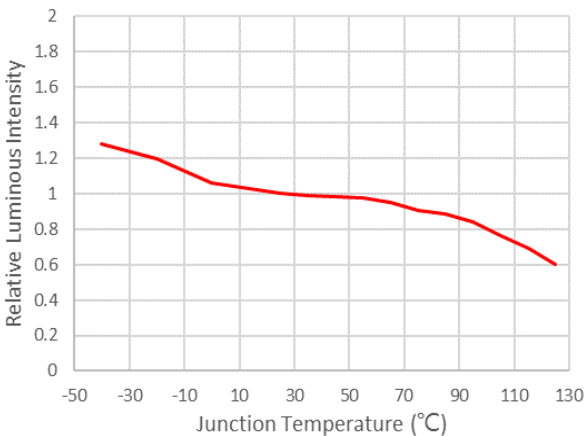
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27. Typical Electrical Optical Characteristics Curves

White (Non-Linear Compensation)



Red (Non-Linear Compensation)

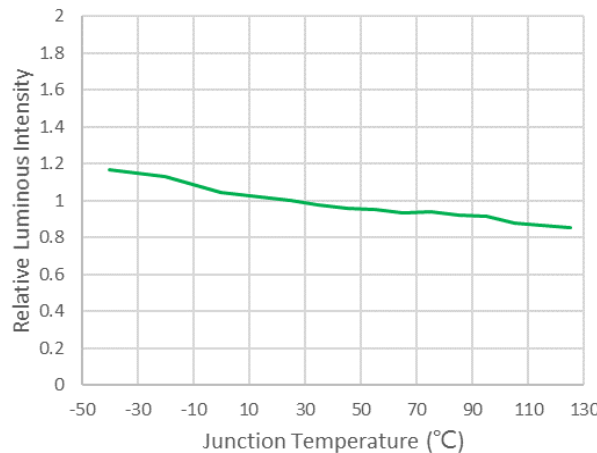


TBD

Green

Relative Luminous Intensity vs Junction Temperature

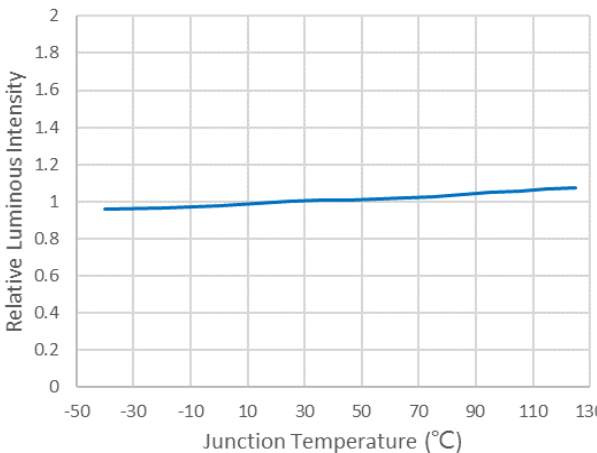
T_J = 25°C ; Color Set Point = (0, 255, 0)



Blue

Relative Luminous Intensity vs Junction Temperature

T_J = 25°C ; Color Set Point = (0, 0, 255)



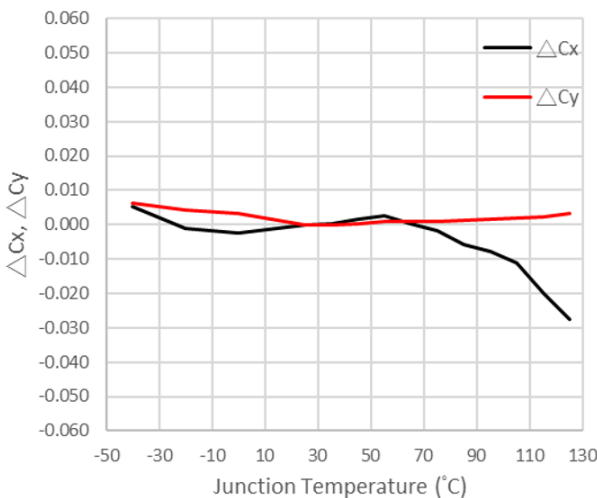
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28. Typical Electrical Optical Characteristics Curves

Chromaticity Coordinate Shift vs
Junction Temperature

White (Non-Linear Compensation)

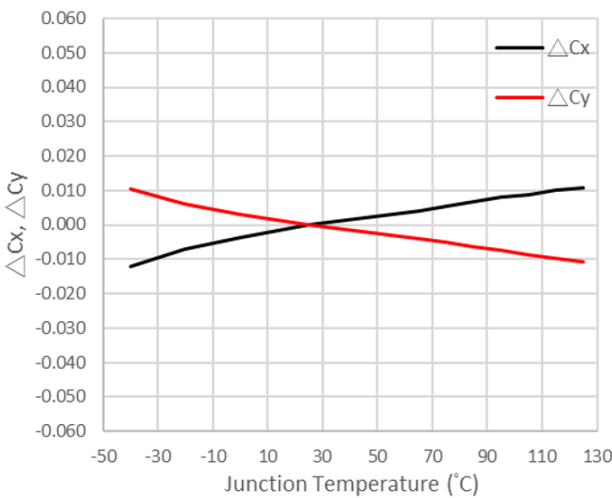
$\Delta Cx, \Delta Cy (250C)=f(Tj)$; Color Set Point=(255,255,255)



Relative Luminous Intensity vs
Junction Temperature

Red (Non-Linear Compensation)

$\Delta Cx, \Delta Cy (250C)=f(Tj)$; Color Set Point=(255, 0, 0)

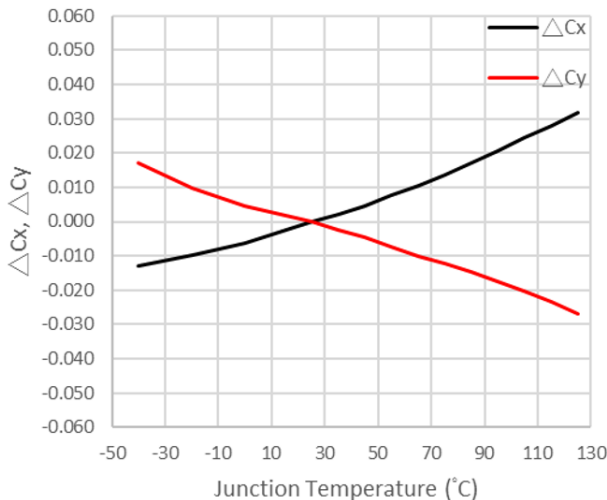


TBD

Relative Luminous Intensity vs
Junction Temperature

Green (Non-Linear Compensation)

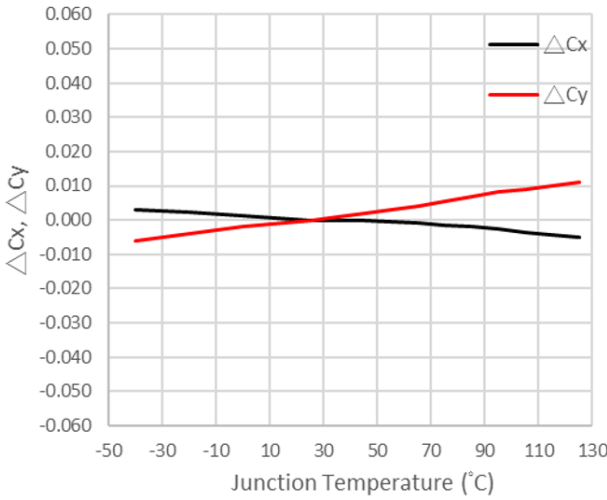
$\Delta Cx, \Delta Cy (250C)=f(Tj)$; Color Set Point=(0, 255, 0)



Relative Luminous Intensity vs
Junction Temperature

Blue (Non-Linear Compensation)

$\Delta Cx, \Delta Cy (250C)=f(Tj)$; Color Set Point=(0, 0, 255)



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29. Soldering Temperature Reflow Profile

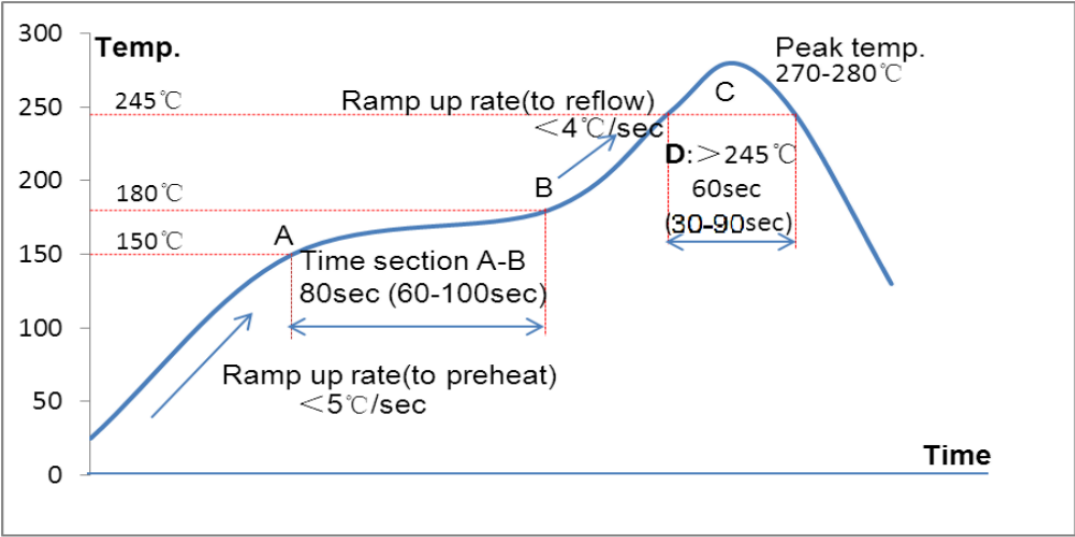


Fig 41. Reflow Soldering information

TBD					
Parameter		Recommendation	Lower limit	Upper limit	Unit
A	Soak zone start	150	140	160	°C
B	Soak zone end	180	160	200	°C
A-B	Section time	80	60	100	S
C	Peak temperature	240	230	255	°C
D	Time above 220°C	50	30	90	S

Table 30. Reflow Soldering Parameter

Note:

The above temperature curve is considered to be a recommended technology one for reference. It can be adjusted due to actual technology demand.

We do not provide explicit or implied guarantees for the accuracy of the data. We are not liable for any loss caused by the use of this information or specified materials

30. Handling Precautions

30-1. Handling precautions

1.Storage

- It is recommended to store the products in the following conditions:
 - Humidity: 10% ~ 55% R.H.
 - Temperature : 18°C~28°C(64.4°F~82.4°F)
- Shelf life in sealed bag: 12 month at < 18°C~28°C and < 10% ~ 55% R.H.
- After the package is Opened, the products should be used within 1 weeks or they should be stored in a vacuumed and sealed with silica gel inside at 30°C or less and 20% R.H. or less.
- After opening the package, the LEDs should be kept at 30°C or less and 20%RH or less.
- When restoring unused the LEDs with anti-electrostatic bag, seal off the anti-electrostatic bag so that no gas and humidity can get it.
- When storage the LEDs in the corrugated cardboard, it may make the LEDs discolored because of minute sulfur gas from the corrugated cardboard.
- Do not use over 10 days in case of using corrugated cardboard.
- Recommend corrugated cardboard containing sulfur less than 850ppm when It is Inevitable use of corrugated cardboard.
- Recommend using material type of PP or PET tray to storage the PCBs or assemblies containing the LEDs. and Insert the silica gel into each of tray.
- Use the anti-electrostatic box with anti-electrostatic cover to prevent Volatile Organic Compounds, sulfur gas and humidity when storing the bundle of the PCBs or assemblies containing the LEDs.
- Do not use bubble wrap directly on top of LEDs. It may cause damage the LEDs
- Do not use rubber band.
- The LEDs must be stored in a clean environment.
- Do not expose the LEDs to direct sunlight.
- Before opening the package, the LEDs should be kept at 18°C~28°C or less and 10% ~ 55% RH or less.
- The LEDs should be used within a year.

30-2. Handling precautions

2.Baking

- Suggest packing open after 1 weeks, before use baking products, conditions as follows:
 - ① $60\pm 3^{\circ}\text{C}$ X 6hrs and $< 5\%\text{RH}$, for reel .
 - ② $125\pm 3^{\circ}\text{C}$ X 2hrs, for single LED
- It shall be normal to see slight color fading of carrier (light yellow) after baking in process.

3.Heat Generation

- When the LEDs are illuminating, operating current should be decided after being considering the ambient maximum temperature.
- Please consider the heat generation of the LED when it is designed the PCB.

4.Cleaning

- If user needs cleaning of the LEDs, use of Isopropyl Alcohol or Ethylene Alcohol is recommended in 5 minutes at room temperature. and dry at room temperature for 15 minutes before use the LEDs. If user uses other than Isopropyl Alcohol or Ethylene Alcohol as cleaning material, it should be not dissolve the LEDs.
- Freon solvents should not be used to clean the LEDs because of worldwide regulations.

5.Static Electricity

- Static electricity or surge voltage damages the Power SMD . It is recommended that a wrist band or an anti-electrostatic glove be used when handling the LEDs.
- A tip soldering iron is requested to be grounded. An ionizer should also be installed where risk of static.
- All devices, equipment and machinery must be properly grounded (via $1\text{M}\Omega$).
- If the LEDs is applied at voltage over maximum value, it may cause damage or destruction of the LEDs
- The LEDs damaged or destructed by anything may cause an increase in leak current, lowered turn on forward voltage, or the LEDs at low forward current.
- It is important to eliminate the possibility of surge current when user designs circuit.

30-2. Handling precautions

6. Product Warranty and Quality Conditions

- The warranty period for this product is one (1) year from the date of shipment.
- Product manual versions may be modified without prior notification. Customers are responsible for obtaining the most recent version of information prior to ordering and must verify that the information is complete and up-to-date.
- All semiconductor products may fail or malfunction under specific conditions. When using this product, the purchaser must comply with safe usage standards and take necessary safety precautions to prevent potential risks that could cause personal injury or property damage.

30-3. Handling LED

- User's working and testing environment can be a important factor to discolor. because unclean testing chamber or working place with wet floor may cause discolor of the LEDs.
- When handling the LED with tools like tweezers or nipper, do not apply Mechanical Forces directly on LED's Surface.
- Do not directly touch LED's surface with hand . It may contaminate the surface and affect on optical characteristics.
- The LEDs should be handled from side because LED's molding material may be damaged with scratching on surface, piercing molding material and broking wire.
- When doing surface mounting technology, If the encapsulation material of the LEDs is silicone, do not any stress or pressure that may cause that emitting surface area of the LEDs resin can be cut, chipped, delaminate or deformed, causing wire bonding breaks and destruction of the LEDs.
- Recommended that the picking up nozzle optimize wider than emitting surface area of the LEDs and setting so that it does not damage the silicone resin.
- Maintain cleanness of picking up nozzle.
- Dropping the LEDs may cause damage.
- Do not contaminate emitting surface area of the LEDs.

31. Dimension of Tape / Reel

31-1. Tape Dimension

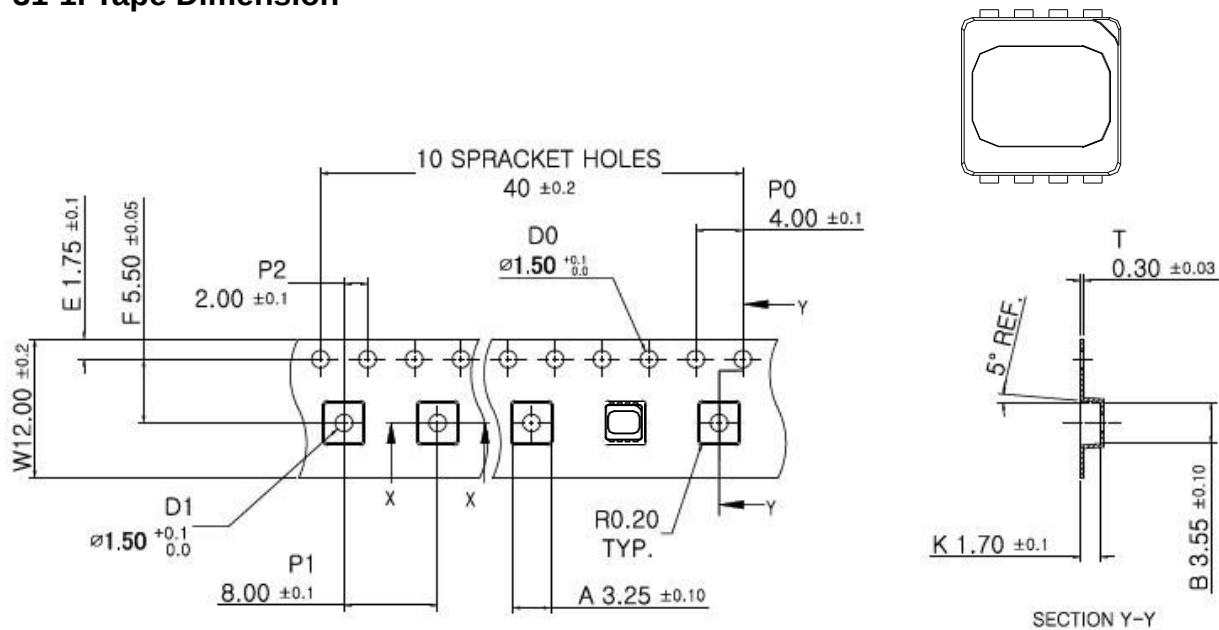


Fig 42. Tape Dimension

31-2. Reel Dimension

TBD

Direction of Feed

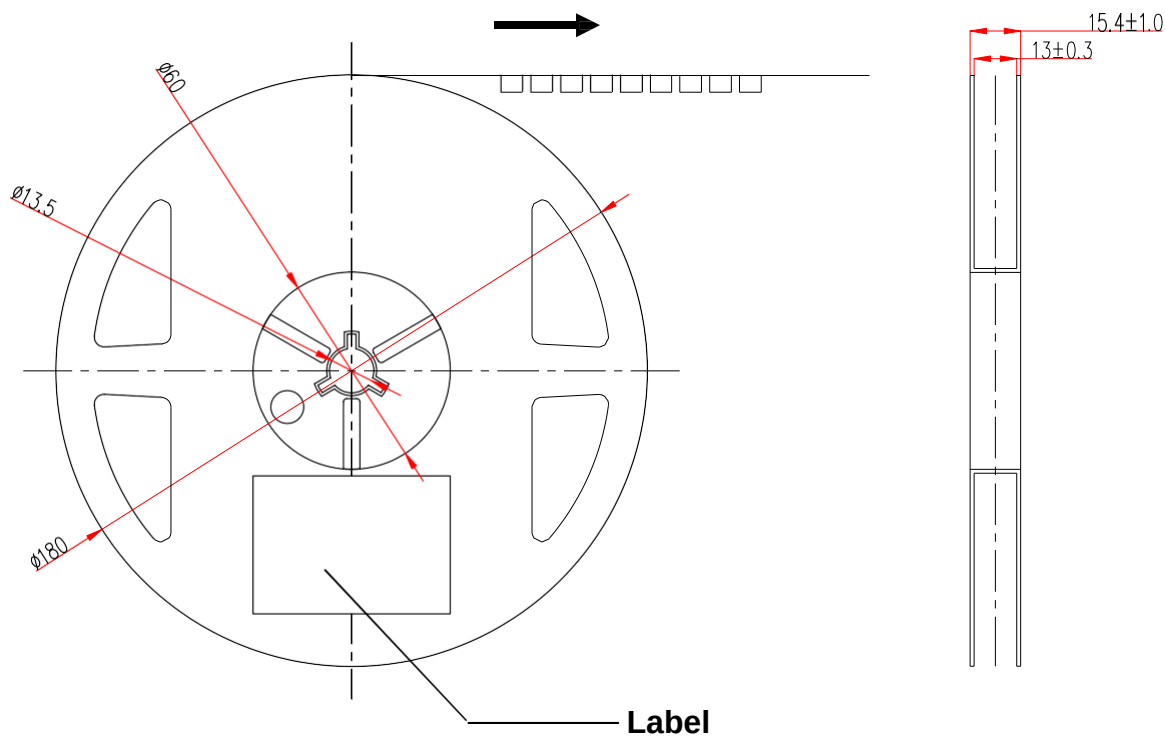


Fig 43. Reel Dimension

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32. Packing Dimension

Unit :mm

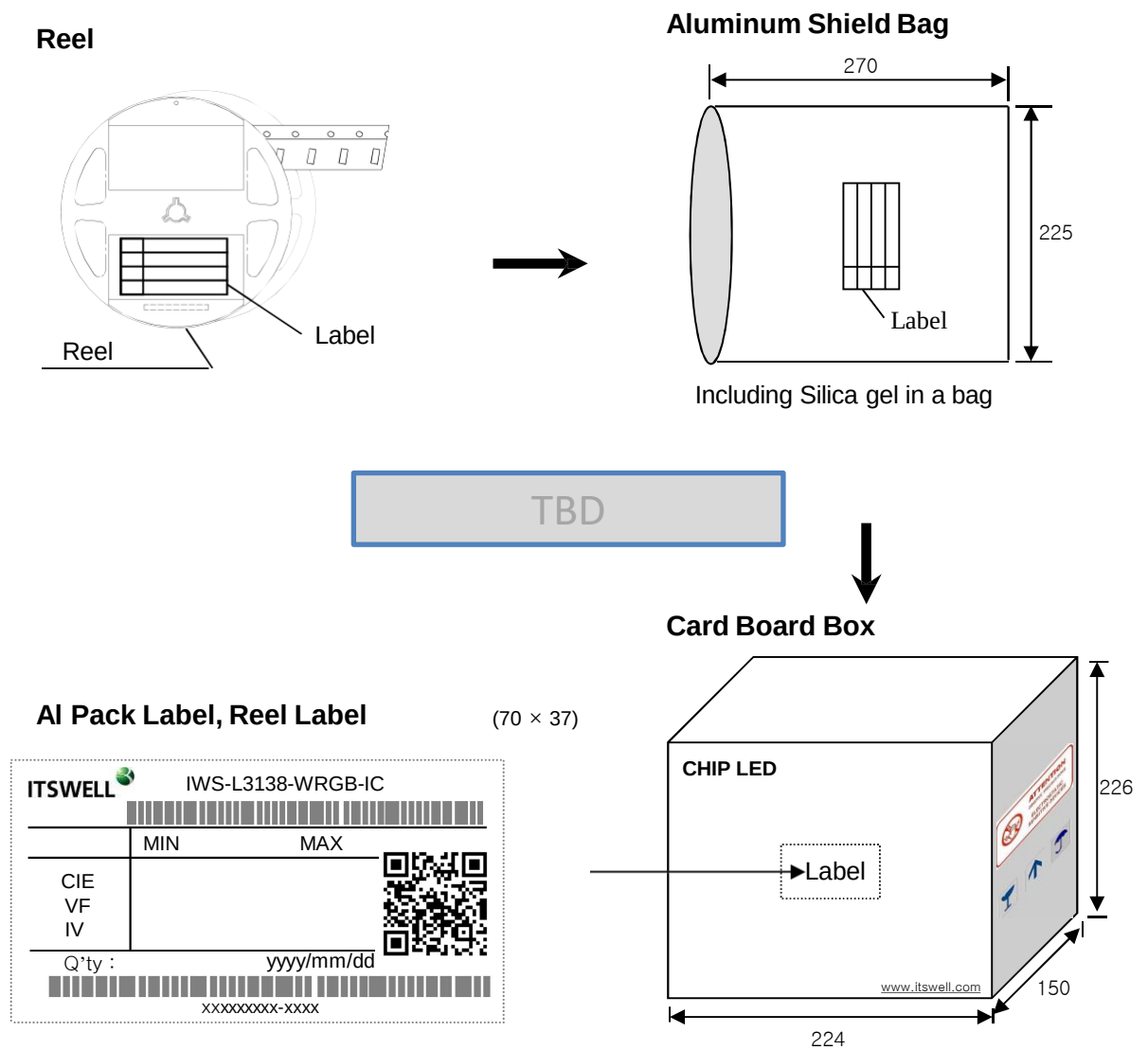


Fig 44. Packing Dimension

	Dimensions (mm)	Reel / Box	Q'ty / Box(pcs)
Reel	Φ180mm, 15mm Width	—	~ 1,000 Max
Al Shield Bag	270x225	—	~ 1,000 Max
Card board Box	224 x 150 x 226	8 Max	8,000 Max

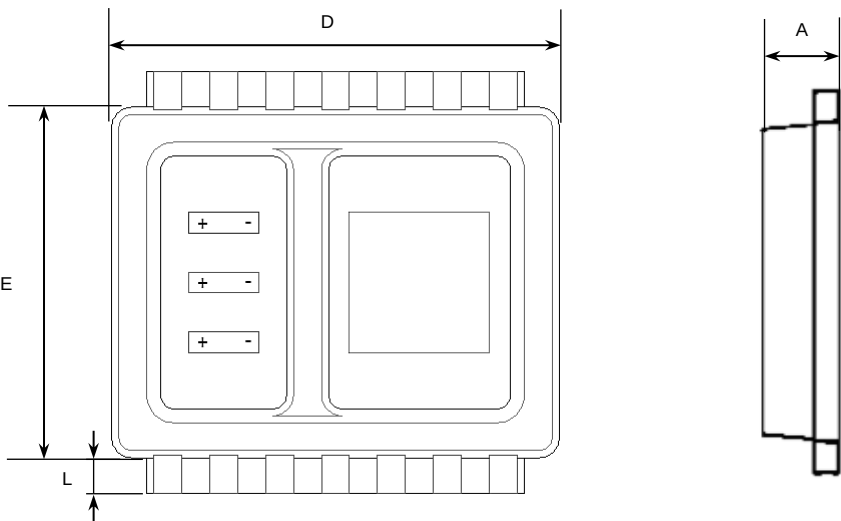
Table 26. Packing Dimension

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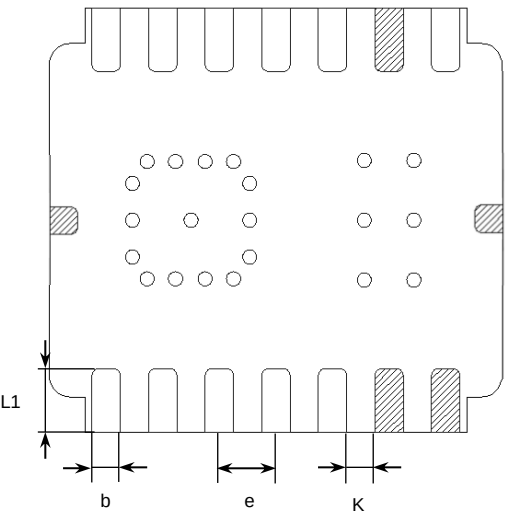
33. Physical Dimensions

DFN 3225-14L Package Outline Dimension

Unit :mm



TBD



Symbol	Dimensions in mm		
	Min	Nom	Max
A		0.60	
b	0.20 BSC		
D		3.20	
E		2.50	
e	0.40 BSC		
K	0.20 BSC		
L		0.30	
L1		0.45	

Fig 45. Package Outline

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34. Recommended Footprint

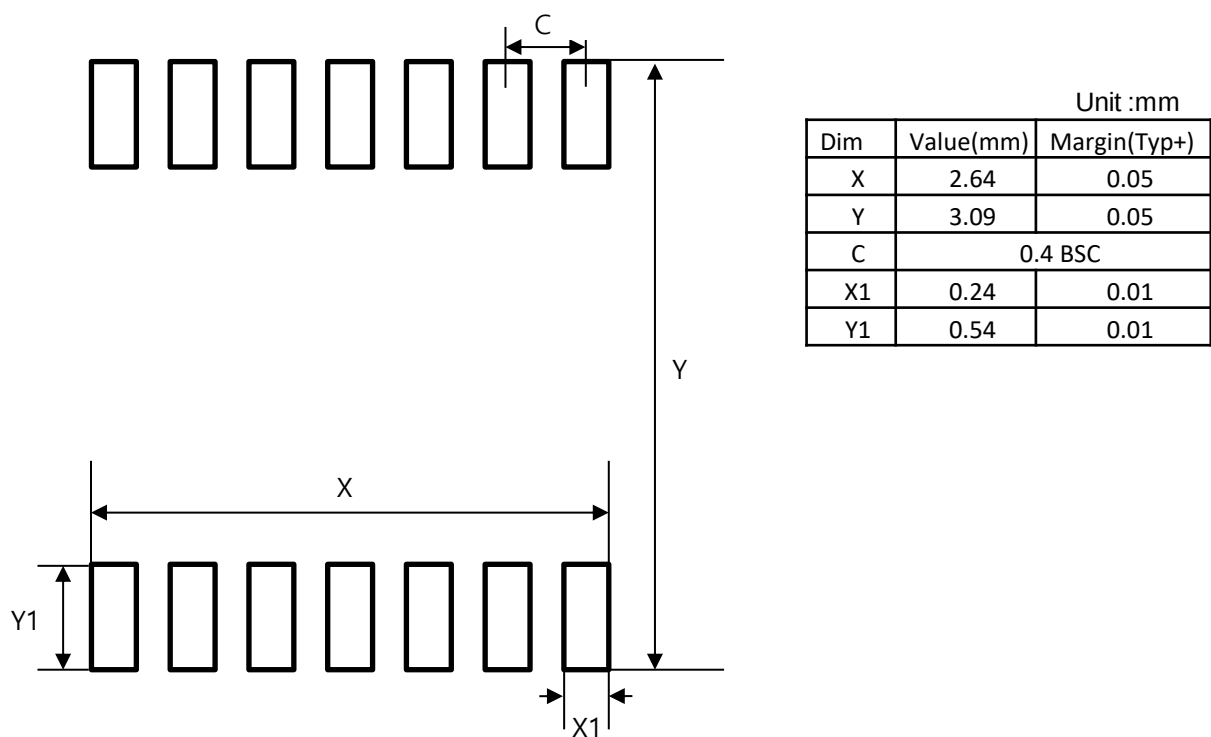


Fig 46. Suggested Footprint

Note: The reflow peak soldering temperature is peak 260°C and it is specified based on IPC/JEDEC J-STD-020 “Moisture/Reflow Sensitivity Classification for Non-Hermetic Solid State Surface Mount Devices”

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